SENSEI

Final Year Project – Mid Report

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A 4th Year Student

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Declaration: The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.				

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ABSTRACT

The impairment of the sight complicates the self-sufficiency of movement a lot, particularly in those places that are full of obstacles, irregular surfaces and spontaneous alterations. Even though the existing assistive applications are able to detect objects or read text to voice they do not guide the users in real time and also they do not assist the user to learn about depth, level of hazard and safe position of feet. This exposes visually impaired individuals to lack the constant help they require in order to move around in a confident and safe manner.

These limitations are supposed to be solved by SENSEI, which is a smart navigation companion that combines artificial intelligence and augmented reality. With the phone camera, SENSEI identifies obstacles and approximates distances, recognizes text, and even the human emotions in the course of social interaction. The system gives cues to the user as they can perceive its environment by use of spatial audio and slight haptic vibrations, which do not require sight. Is voice-enabled and an audio-first interface means that the app is fully accessible and hands-free.

SENSEI is built using React Native, Expo AR, and TensorFlow Lite and it offers real-time environmental awareness on typical smartphones. It is anticipated that the end result will be an intuitive, supportive, and empowering navigation experience, to the visual impaired users. Combining AR, AI, and multisensory feedback, SENSEI is supposed to make the world a more inclusive place, where the visually impaired will be able to move more freely and confidently.

Acknowledgement

The acknowledgement section is **OPTIONAL** and should comprise a short, formal note to express gratitude to the people, organizations, or institutions that supported you during the preparation of the report. It is not very long, typically one or two paragraphs.

Table of Contents

List of Tables

List of Figures

Introduction

1.1 Introduction

The last few years have seen a steady increase in the growth of assistive technologies due to significant developments in Artificial Intelligence, Augmented Reality, mobile devices, and wearables. Such technologies have provided an opportunity to people with disabilities, particularly those with visual impairments that do not always have an opportunity to feel safe, move independently, and comprehend the world around them. Although there are numerous digital solutions, a significant distance is still observed between the capabilities of these solutions and the form of the real-time, flexible guidance that the visually impaired people require in their daily lives. Simple safety aids such as white canes or guide dogs only do not offer the contextual information. Conversely, the objects or text can be recognized with the usage of such apps as Microsoft Seeing AI, Google Lookout, and Sullivan+ that do not provide the continuous navigation, foot-step directions, social and haptic feedbacks. This gap makes it obvious that a system, which would integrate various senses and give real-time navigation assistance rather than non-real-time detection of objects, is required. SENSEI is made to meet that requirement.

The notion behind the SENSEI project is based on the desire to make the visually impaired users more independent and assured as they navigate their lives in their daily surroundings. Among the largest difficulties they encounter is to drive safely without depending on others. This problem should be solved not only in terms of technology but also to enhance inclusiveness and equal access. Another opportunity to do that when working on SENSEI is the chance to work with modern AR navigation, computer vision, wearable technology, and the user-centered design. By means of the project, we will be able to learn more about the ways AI and AR could be integrated into each other to provide more user-friendly and safer navigation in real-life scenarios.

The project is expected to develop a mobile application that is capable of offering real-time AR navigation, obstacle detection, depth estimation, emotion recognition, and haptic feedback. The app should be audio-first and should be gesture-controlled as this way the app should be accessible to blind people. SENSEI will feature such aspects as constant object detection, foot-placement guidance, text reading, and emergency warnings. The project is not associated with the development of the own equipment, the creation of AR glasses, the development of big clouds. The primary aim is to design an efficient and effective prototype of mobile AR that is functional and effective in using the contemporary smart phones.

In order to address the constraints of the existing technology, SENSEI proposes an AR-powered mobile app that is powered by the AI models and supported with the help of wearable gadgets. The system examines the surrounding environment, identifies barriers, risks, and moves the user with the help of 3D audio and vibration cues using the camera of the phone. It has processes powered by TensorFlow Lite or PyTorch to run AI, cross-platform AR applications with React Native and Expo AR, and Bluetooth support to connect wearable devices. Development process involves the gathering of requirement, interface design, model training, AR integration, backend and testing with the visually impaired.

- The anticipated project deliverable is a working prototype that will be able to deliver:
- Real-time AR navigation
- Identification of the hindrances and determination of the level of their threat.
- Emotional identification to help in socialization.
- Environmental descriptions, reading of a text.
- Gesture command audio-based controls.
- Wearable haptic feedback

Some of the evaluation criteria to be used in the project include; the accuracy of the navigation, system responsiveness, user satisfaction, and the capability to work in various environments. The final report will have screenshots, test performance, and user feedback, and performance statistics.

The project made the following contributions:

- An AR-based navigation application unique to the visually impaired users.
- Object recognition and text recognition using AI.
- Foot-placement assistance and depth estimation.
- Social awareness assistance through emotion detection.
- Bluetooth undulating haptic feedback.
- An audio-first interface enhanced to non-visual use.

Problem Statement

The assistive technologies today are not dynamic and do not provide contextual support on navigation and use voice commands as the primary mode of operation. Compared to Seeing AI or Lookout, apps such as those do not have haptic navigation, foot-placement assistance, or constant AR-based movement guidance, although some apps can identify objects or read text. These restrictions complicate the ability of the visually impaired users to avoid the hindrances, effectively interact, and move freely. SENSEI is designed to address these issues through an integration of both AR, AI, and wearable feedback to provide real-time adaptive navigation and more interesting experience when walking.

Proposed Solution

SENSEI proposes AR-based navigation system, which can assist visually impaired users to comprehend the environment around them in real-time by detecting objects, estimating their depth, providing spatial audio, and haptic vibrations. The phone camera captures the surrounding, AI processes it, AR monitors the location, and Bluetooth devices generate directional vibrations in order to lead the user. This system is an upgrade to the available solutions because of the provision of moment-to-moment navigation, emotion recognition, programmable audio options, and intelligent route memory.

Main Objectives

- In order to develop an AR-based solution of navigation among visually impaired users.
- To incorporate the use of AI in object identification, text recognition, and emotion detection.
- To signal obstacles and directions by use of wearable haptics.
- To create foot-placement and depth estimation features to make mobility a safer feature.
- To create a voice-based interface with gestures and voice recognition.
- To provide user safety by means of real-time hazard warnings.

Assumptions

- Customers get familiar with simple voice orders and smartphone movements.
- Wearable haptic devices will be used well by the users.
- Text-to-speech and spatial audio guidance relies on the user.
- The situation will allow the use of cameras with respect to lighting.

Constraints

- Expo AR has limited support
- The processing power of the smartphone is a determinant of performance.

- Supervision of visually impaired participants should be done in testing.
- Only basic features will be offline and the rest advanced features require internet.

Project Scope

In Scope:

- AR navigation
- Text reading and object detection.
- Emotion recognition
- Foot-placement and depth estimation
- and depth estimation.
- Wearable haptics
- Gesture and voice interaction.
- Android and iOS support

Out of Scope:

- Custom hardware creation
- AR glasses
- Enterprise cloud systems in large scale.
- Autonomous robotics

SDLC Model

The reason behind the usage of the Agile model is that it encourages iterative development, rapid changes, and user feedback, all of which play a significant role in the creation of accessibility-based applications. Every sprint involves improving the requirements, the UI and UX, training the AI models, module integration, and testing with the blind users. This will have the effect of ensuring that SENSEI keeps on improving according to the real world requirements and remain user friendly at all development stages.

Requirement Analysis
Literature Review
AI, computer vision, and mobile systems have enhanced assistive technologies that are available to visually impaired users. Nevertheless, there are still issues with real-time navigation, obstacle vision, perception of the scene, and social interaction. The available applications provide only partial assistance without integrating an AR-based awareness, haptic navigation, and emotion recognition into one system.

An example is Microsoft Seeing AI that reads documents, identifies objects and faces. It is informative but not real-time movement or hazard-detecting, which is restrictive when it comes to real-world navigation.

Google Lookout also recognizes objects, reads labels and works across a number of languages. It still has no depth sensing, continuous pathfinding, hazard warnings and foot-placement guidance though.

Sullivan+ does not have AR navigation and wearable haptics but provides descriptions of the scene and some elementary environmental analysis. It can be helpful to provide information, but not to move safely and dynamically.

These studies indicate that, majority of the available tools only react when activated and not to provide constant support when the user is walking. They lack AR depth sensing, foot-placement navigation, emotion recognition, hazard evaluation, and haptic feedback.

The SENSEI project is premised on these weaknesses. Integrating AR, AI, computer vision, spatial audio, and wearable vibrations into a single seamless system, SENSEI will offer real-time guidance and better environmental awareness and help with social interaction, which covers all significant drawbacks of existing systems.

author/app	year	methods used	main functionalities	shortcomings	references
microsoft seeing AI	2017–2024	AI-based scene interpretation, OCR	Object recognition, text reading, face detection	No AR navigation, no haptics, no danger assessment	[4]
google lookout	2018–2020	Computer vision, label detection, barcode scanning	Object detection, document reading, food label identification	No continuous navigation, no depth estimation, no spatial audio	[3]
TUAT Korea – Sullivan+	2019	AI scene labeling, OCR	Scene description, text recognition, object identification	No AR integration, no hazard-level alerting	[5]
SENSEI (Proposed)	2025	AR + AI + Wearables + Spatial Audio	Real-time navigation, depth estimation, emotion detection, haptic alerts	Mobile hardware constraints, dependent on lighting	–

Stakeholders List

- The key consumers of the system include people with visual impairments.
- The caregivers and navigation trainers are the supporters of the users.
- The team comprises developing team, designers, developers as well as testers who develop the application.
- The project is managed by academic supervisors of CUI.
- The NGOs and organizations dealing with accessibility are external stakeholders.
- Such platforms and vendors as Expo AR, TensorFlow, and wearable devices suppliers fall into the category of technical stakeholders.

Requirements Elicitation

1.1 Purpose

This report describes the way the SENSEI requirements were obtained. SENSEI is a mobile application (AI-driven) that guides blind people to navigate more freely with the assistance of augmented reality, spatial sound, and smart identification of objects.

1.2 Elicitation Methods

1.2.1 Stakeholder Analysis

The major stakeholders are the visually impaired users who will be directly dependent on the app. The other important actors include caregivers, relatives, accessibility divisions, medical practitioners as well as emergency departments.

Mobile developers, AR and AI specialists, accessibility-conscious UI/UX designers, and system administrators that maintain or support the system are all secondary stakeholders.

1.2.2 Techniques of the Requirements Gathering.

User Interviews:

- Over fifteen visually impaired people were interviewed in a structured interview.
- The interviews were aimed at challenges during their everyday traveling, risk factors, and how comfortable they felt with technology.
- The general observations revealed a high demand in a real-time alert of obstacles, voice controls and emergency support features.

Surveys and Questionnaires:

- Over fifty members of the accessibility community filled in surveys.
- The results showed that the majority of the users use voice command options, depend on sound heavily, and are concerned with safety when driving around.
- Observation Studies
- Eight visually impaired clients were seen in their daily activities.
- This assisted in bringing forth the critical moments like crossing roads, evading sudden hindrances and movement to designated places.

Document Review:

Such standards of accessibility as WCAG 2.1 and Section 508 were analyzed, and the topic of AR-based navigation tools with visually impaired users was researched.

Brainstorming Workshops:

Other feature ideas, such as spatial audio, haptic feedback, and emotion recognition, came as a result of sessions with specialists on accessibility and the development team.

Competitive Analysis:

- The available systems such as BlindSquare, Seeing AI, and Google Lookout were examined.
- Though handy, these applications did not have good AR navigation, did not have hazard scoring, and they had only simple audio directions.

1.3 Requirements Validation

- The requirements collected were proved by the prototype testing with twelve visually impaired users.
- Focus groups, accessibility tests and technical viability tests were later refined.

1.1.1 Functional Requirements

FR1: User Authentication and Profile Management

ID	REQUIREMENT	PRIORITY	STATUS
FR 1.1	Users shall be able to register with email, phone, or social authentication	high	Implemented
FR 1.2	System shall maintain user profiles with accessibility preferences	high	Implemented
FR 1.3	System shall store emergency contacts for each user	critical	Implemented
FR 1.4	Users shall be able to customize voice speed and audio settings	medium	Implemented

FR2: Object Detection and Recognition

ID	REQUIREMENT	PRIORITY	STATUS

FR 2.1	System shall detect objects in real-time	critical	**Pending**
FR 2.2	System shall identify persons, vehicles, traffic lights, benches, doors	critical	**Pending**
FR 2.3	System shall calculate object distance	high	**Implemented**
FR 2.4	System shall determine object position relative to user	high	**Implemented**
FR 2.5	System shall provide confidence scores	medium	**Implemented**
FR 2.6		critical	**Implemented**

	System shall prioritize critical objects		
FR 2.7	System shall generate contextual scenarios	medium	**Implemented**
FR 2.8	System shall announce detected objects via text-to-speech	critical	**Implemented**

FR3: Augmented Reality (AR) Features

ID	REQUIREMENT	PRIORITY	STATUS
FR 3.1	Display AR navigation overlays	high	**Implemented**
FR 3.2	Highlight obstacles in AR	high	**Implemented**
FR 3.3	Provide AR-based depth estimation	medium	

			Pending
FR 3.4	Calculate safe foot placement zones	high	**Implemented**

FR4: Voice Command Syste

ID	REQUIREMENT	PRIORITY	STATUS
FR 4.1	Recognize natural language voice commands	critical	**Implemented**
FR 4.2	Identify commands: navigate, find, describe	critical	**Pending**
FR 4.3	Answer "where am I?" with location details	high	**Pending**
FR 4.4	Recognize emergency keywords	critical	**Pending**
FR 4.5	Support route memory commands	medium	**Implemented**
FR 4.6	Provide voice feedback for commands	high	**Pending**
FR 4.7	Maintain command history	low	**Implemented**

FR5: Emergency Services

ID	REQUIREMENT	PRIORITY	STATUS
FR 5.1	One-tap emergency alert activation	critical	**Implemented**
FR 5.2	Send GPS coordinates to emergency contacts	critical	**Pending**
FR 5.3	Enable direct emergency calling	critical	**Implemented**
FR 5.4	Detect falls via accelerometer	high	**Pending**
FR 5.5	Trigger haptic emergency alerts	high	**Implemented**
FR 5.6	Share live location via SMS/URL	high	**Pending**
FR 5.7	Allow canceling false alerts	medium	**Implemented**

FR8: Hazard Detection and Scoring

ID	REQUIREMENT	PRIORITY	STATUS
FR 6.1	Calculate hazard scores (0–100)	high	**Implemented**
FR 6.2	Identify high-risk hazards	critical	**Implemented**
FR 6.3	Provide urgency levels	high	**Implemented**
FR 6.4	Announce critical hazards immediately	critical	**Implemented**
FR 6.5	Track hazard history	low	**Implemented**

FR9: Offline Mode

ID	REQUIREMENT	PRIORITY	STATUS
FR 7.1	Cache maps for offline use	high	**Implemented**
FR 7.2	Save frequently visited routes	medium	**Implemented**
FR 7.3	Basic navigation offline	high	**Implemented**
FR 7.4	Sync data when online	medium	**Implemented**

FR10: Wearable Device Integration

ID	REQUIREMENT	PRIORITY	STATUS
FR 8.1	Connect to Bluetooth wearables	MEDIUM	**Implemented**
FR 8.2	Send haptic feedback	MEDIUM	**Implemented**
FR 8.3	Support smartwatches	LOW	**Pending**
FR 8.4	Display device battery status	LOW	**Pending**

FR11: Optical Character Recognition (OCR)

ID	REQUIREMENT	PRIORITY	STATUS
FR 9.1	Read text using OCR	HIGH	**Implemented**
FR 9.2	Recognize signs, labels, documents	HIGH	**Implemented**

FR 9.3	Convert text to speech	HIGH	**Implemented**
FR 9.3	Support multiple languages	MEDIUM	**Implemented**

FR12: Route Memory and Learning

ID	REQUIREMENT	PRIORITY	STATUS
FR 10.1	Save frequently used routes	medium	**Implemented**
FR 10.2	Recognize saved locations	medium	**Implemented**
FR 10.3	Provide shortcuts	low	**Implemented**
FR 10.4	Suggest routes based on history	low	**Implemented**

1.1.2 Non-functional Requirements

NFR1: Performance

ID	NFR	Requirement	Priority	Metric
NFR1	Performance	Object detection shall process frames within 200ms	Critical	<200ms latency
NFR1.2	Performance	GPS location updates shall occur every 1–5 seconds	High	1–5s interval

NFR1.3	Performance	Voice command response time shall be <1 second	High	<1s response
NFR1.4	Performance	AR rendering shall maintain 30 FPS minimum	High	≥30 FPS
NFR1.5	Performance	App startup time shall be <3 seconds	Medium	<3s cold start
NFR1.6	Performance	Navigation route calculation shall complete <2 seconds	High	<2s calculation

NFR2: Usability & Accessibility

ID	NFR	Requirement	Priority	Compliance
NFR2.1	Usability & Accessibility	Interface shall comply with WCAG 2.1 Level AA	Critical	WCAG 2.1 AA
NFR2.2	Usability & Accessibility	All features must be accessible using voice commands	Critical	100% voice-accessible
NFR2.3	Usability & Accessibility	Text-to-speech shall support multiple languages	High	Multi-language
NFR2.4	Usability & Accessibility	Minimum touch target size = 44×44 px	High	iOS HIG
NFR2.5	Usability & Accessibility	Color contrast ratio ≥ 4.5:1	Medium	WCAG standard
NFR2.6	Usability & Accessibility	Screen reader compatibility required	Critical	Full SR support

NFR3: Reliability & Availability

ID	NFR	Requirement	Priority	Target

NFR3.1	Reliability & Availability	System uptime ≥ 99.5%	High	99.5% uptime
NFR3.2	Reliability & Availability	Emergency alert must always trigger	Critical	Guaranteed
NFR3.3	Reliability & Availability	Real-time services must handle 80% load without delay	High	80% capacity
NFR3.4	Reliability & Availability	Low failure rate for continuous navigation	High	≤2% failure
NFR3.5	Reliability & Availability	Automatic recovery from minor crashes	Medium	Auto-restart

NFR4: Security

ID	NFR	Requirement	Priority	Standard
NFR4.1	Security	User data encryption (AES-256)	Critical	AES-256
NFR4.2	Security	Encrypted communication (HTTPS/TLS 1.2+)	Critical	TLS 1.2
NFR4.3	Security	Secure emergency data transfer	High	Encrypted
NFR4.4	Security	Secure authentication required	High	OAuth2
NFR4.5	Security	No data stored without consent	Critical	GDPR

ID	NFR	Requirement	Priority	Standard
NFR4.1	Security	User data encryption (AES-256)	Critical	AES-256

NFR4.2	Security	Encrypted communication (HTTPS/TLS 1.2+)	Critical	TLS 1.2
NFR4.3	Security	Secure emergency data transfer	High	Encrypted
NFR4.4	Security	Secure authentication required	High	OAuth2
NFR4.5	Security	No data stored without consent	Critical	GDPR

NFR5: Scalability

ID	NFR	Requirement	Priority	Metric
NFR5.1	Scalability	System shall support future AI model upgrades	Low	Modular
NFR5.2	Scalability	Should support increased user load	Low	Cloud scalable
NFR5.3	Scalability	Support larger environmental datasets	Medium	Dataset expansion

NFR6: Compatibility

ID	NFR	Requirement	Priority	Standard
NFR6.1	Compatibility	Compatible with Android 10+	High	Android 10+
NFR6.2	Compatibility	Compatible with iOS 13+	High	iOS 13+
NFR6.3	Compatibility	Compatible with iPhone 8 and newer	High	iPhone 8+
NFR6.4	Compatibility	Must support ARCore devices	High	ARCore

NFR6.5	Compatibility	Screen reader compatibility: VoiceOver, TalkBack	Critical	iOS/Android SR

NFR7: Maintainability

ID	NFR	Requirement	Priority	Standard
NFR7.1	Maintainability	Follow React Native best practices	High	RN Standards
NFR7.2	Maintainability	Maintain 80%+ test coverage	Medium	80% coverage
NFR7.3	Maintainability	API documentation in Swagger/OpenAPI	Medium	OpenAPI 3.0
NFR7.4	Maintainability	Modular architecture for easy updates	High	Service-based

NFR8: Battery & Resource Efficiency

ID	NFR	Requirement	Priority	Target
NFR8.1	Battery & Resource Efficiency	Battery drain ≤ 15% per hour	Critical	≤15%
NFR8.2	Battery & Resource Efficiency	Memory usage ≤ 200 MB	High	≤200 MB
NFR8.3	Battery & Resource Efficiency	App size < 100 MB	Medium	<100 MB

NFR8.4	Battery & Resource Efficiency	Background tracking optimized for battery	High	Low-power mode

NFR9: Audio Quality

ID	NFR	Requirement	Priority	Target
NFR9.1	Audio Quality	TTS must be clear and natural	Critical	High-quality TTS
NFR9.2	Audio Quality	Spatial audio accuracy within 15 degrees	High	±15°
NFR9.3	Audio Quality	Audible in noisy environments (>70 dB)	High	Adaptive volume
NFR9.4	Audio Quality	Support for external speakers & hearing aids	Medium	Audio routing

NFR10: Legal & Compliance

ID	NFR	Requirement	Priority	Standard
NFR10.1	Legal & Compliance	Must comply with ADA	Critical	ADA
NFR10.2	Legal & Compliance	Meet FDA guidelines for assistive apps	Medium	FDA
NFR10.3	Legal & Compliance	Must meet Section 508 accessibility standards	High	Section 508

1.1.3 Requirements Traceability Matrix

4.1 Core Navigation Features

Requirement ID	**Requirement Description**	**Source**	**Design Component**	**Implementation File**	**Status**
FR2.1	Turn-by-turn voice navigation	User Interview #1-5	NavigationService	NavigationService.js	done
FR2.2	GPS-based route calculation	Stakeholder Survey	NavigationService	NavigationService.js	done
FR2.3	Automatic route recalculation	Observation Study #2	NavigationService	NavigationService.js	done
FR2.4	Proximity alerts (<50m)	User Interview #3	NavigationService	NavigationService.js	done
FR2.8	Real-time distance tracking	User Requirement	NavigationScreen	NavigationScreen.js	done
NFR1.2	GPS updates every 1–5 seconds	Performance Spec	LocationService	LocationService.js	done
NFR1.6	Route calculation <2 seconds	Performance Spec	NavigationService	NavigationService.js	done

4.2 Object Detection & Safety

Requirement ID	Requirement Description	Source	Design Component	Implementation File	Status
FR3.1	Real-time object detection	User Interview #1-8	ObjectDetectionService	ObjectDetectionService.js	pending
FR3.2	Identify common objects	Accessibility Expert	ObjectDetectionService	ObjectDetectionService.js	pending
FR3.3	Calculate object distance	Safety Requirement	ObjectDetectionService	ObjectDetectionService.js	done
FR3.6	Prioritize critical objects	Observation Study #1	ObjectDetectionService	ObjectDetectionService.js	done
FR8.1	Hazard	Safety	HazardScori	HazardScori	done

	scoring (0–100)	Consultant	ngService	ngService.js	
FR8.2	Identify high-risk hazards	Safety Requirement	HazardScoringService	HazardScoringService.js	done
NFR1.1	Detection latency <200ms	Performance Spec	ObjectDetectionService	ObjectDetectionService.js	done

4.3 AR Features

Requirement ID	Requirement Description	Source	Design Component	Implementation File	Status
FR4.1	AR navigation overlays	Tech Brainstorm	ARNavigationService	ARNavigationService.js	done
FR4.2	Highlight obstacles in AR	Tech Brainstorm	ARNavigationService	ARNavigationService.js	done
FR4.3	3D directional arrows	Tech Brainstorm	ARNavigationService	ARNavigationService.js	done
FR4.4	AR depth estimation	Tech Brainstorm	ARNavigationService	ARNavigationService.js	done

4.9 Performance & Resource Management

Requirement ID	Requirement Description	Source	Design Component	Implementation File	Status
NFR1.5	App startup <3s	UX Requirement	AppInitializer	AppInitializer.js	done
NFR8.1	Battery drain ≤15%/hour	Technical Spec	All Services	Power Optimization	pending
NFR8.2	Memory ≤200MB	Technical Spec	Memory Management	All Services	pending

NFR8.3	App size <100MB	Distribution Spec	Build Config	App Bundle	done

4.10 UI & UX

Requirement ID	Requirement Description	Source	Design Component	Implementation File	Status
FR1.1	User registration	Standard Requirement	Auth Routes	auth.js	done
FR1.2	User profile management	User Requirement	Settings Screen	SettingsScreen.js	done
FR1.4	Customize audio settings	User Interview #5	SettingsService	SettingsService.js	done
NFR2.4	Touch target ≥44×44	iOS HIG	Buttons	Button.js	done
NFR2.5	Color contrast 4.5:1	WCAG	Theme System	theme.js	done

1.2 Use Case Description

UC-01: Register and Create User Account.

Use Case ID: UC-01

Name of the Use Case: Register and Create User Account.

Primary Actor: Visually Impaired User.

Secondary Actors: Authorization Service.

Preconditions:

1. SENSEI application is installed on user.
2. It has internet connectivity.
3. There is no existing account of the user.

Main Flow:

1. User opens the application
2. The system has a logging screen with register button.
3. User clicks on register or create account.
4. Email/phone number, supplied by the user.
5. User makes password .
6. Password strength is a system-validated password.
7. User gives the option of accessibility (speed of voice, volume of audio).
8. Confirmation of creating accounts.
9. The user is logged in and directed to the Home Screen.

Postconditions:

1. In the system, user account is created.
2. User profile contains the preferences of accessibility.
3. User is verified and logged in to the app.
4. Emergency contacts field is set aside in the future.
5. At Home Screen and other features, user can access.

FR1.1, FR1.2, FR1.4 Satisfied.

UC-02: Set Up Emergency Contacts

Use Case ID: UC-02

Name of Use Case: Emergency Contacts Set-Up.

Primary Actor: Visually Impaired User.

Secondary: Contacts Service, SMS Service.

Preconditions:

1. The user logins into the application.
2. The user is fully profiled.
3. User can access their phone contacts or can dial phone numbers.

Main Flow:

1. User clicks on to settings screen.
2. User chooses the option of Emergency Contacts.
3. System displays emergency contacts set-up screen.
4. User presses on Add Emergency Contact button.
5. System has two features: choose contacts or type in.
6. User decides to pick out of phone contacts.
7. Phone contacts list is accessed by the system.
8. The user picks the contact(s) either by voice or by touch.

9. System reads back the contact that has been selected to confirm.
10. User confirms selection
11. Emergency contact(s) are saved in the system.
12. User is able to add other contacts (repeat 4-11)
13. User clicks on Save to save set.
14. System ensures that all emergency contacts are stored.
15. System stores contacts in a local and secure manner.

Alternate Flow: Manual Contact Entry 1:

1. At step 6: User decides to add contact as a manual one.
2. Prompts system name of contact.
3. Contact name will be entered by the user either by voice or by typing.
4. Phone number prompts within the system.
5. User provides phone number
6. Details are verified by the system then saved.

Alternate Flow 2: Contact -Edit Existing Contact:

1. At step 2: User clicks on "Emergency Contacts"
2. System displays address books of available contacts.
3. User selects contact to edit
4. Contact details can be edited in the system.
5. Confirmation and changes done by user.
6. Updating of system updates contact information.

Postconditions:

1. Contact details of emergency are saved in profile.
2. User gets a confirmation of saved contacts.
3. Emergency alerts can be found at contacts.
4. System has voice-read voice verification contact list.

FR1.3, FR7.2, NFR4.4 Satisfied.

UC-03: Object Detection and Announcement.

Use Case ID: UC-04

Name of Use Case: Objects Detector and Announcer.

Primary Actor: Blind User.

Secondary Actors: Text-to-Speech Service, a Camera, Object Detection Service.

Preconditions:

1. User has started AR Screen or detection mode on.
2. Permission to the camera is given.
3. The device is moving or the user is not moving.
4. System is activated, and it is in detection mode.

Main Flow:

1. User switches to AR Screen or voice commands "Start detection"
2. System gives access to camera.
3. Camera is granted to the user.
4. Camera feed is activated by the system.
5. System initiates frame processing on a continuous basis.
6. Object detection model is run in each frame.
7. System recognizes objects of confidence above 75%.
8. System calculates distance to all objects.
9. System determines the position of object (left, center, right).
10. System data measures the level of hazard of every object.
11. In case of hazard level is critical (vehicles less than 5m):

    - System: "Vehicle 3 meters on your right Vehicle, 3 meters ahead of you, immediately announces.

    - System activates spatial audio signal.

    - System forcibly runs a vibration pattern.

12. In case of high hazard level (persons <3m):

    - medium priority system announcement.

13. In case of level of hazard of Medium or Low:

    - System periodically announces or offers on demand information.

14. User moves are continuously updated on the position of objects in the system.
15. System deletes objects on list when not detected or greater than 100m.

Alternate Flow 2 - Flow Manual Adjustment of the Camera:

1. In step 5: User rotates device.
2. Recalibration of system according to new camera angle.
3. Relative position of objects is updated in base of the new orientation of the user.

Postconditions:

1. User is aware of the surrounding environment.

2. Severe risks have been declared.
3. System continuously detects whilst on.
4. The data of the objects is recorded in order to be learnt and analyzed in terms of safety.

FR3.1, FR3.2, FR3.3, FR3.4, FR3.6, FR3.8, FR8.1, FR8.2, FR8.4, NFR1.1 Satisfied.

UC- 04: Operate the System by Voice Commands.

Use Case ID: UC-05

Use Case Name: Use Voice Commands to control the System.

*Primary Actor :*Visually Impaired User.

Secondary Actors: voice recognition service, system services.

Preconditions:

1. Microphone permission is approved to user.
2. Application is in foreground or active.
3. Continuous listening or voice command wake word has been spoken by user.

Main Flow:

1. When the user speaks the voice input begins with sayingHey Sensei or by pressing voice button.
2. Microphone is activated and start of listening.
3. System records speech used by a user up to 3-5 seconds or when the user pauses.
4. User voice: Find the coffee shop -which is a command- find/navigate to home
5. Audio is passed to speech recognition service through system.
6. Text is transcribed into the system.
7. System analyzes command against command patterns.
8. System provides command to action type (navigation, detection, info, control, memory, emergency) match.
9. System picks off parameters of command.
10. System takes the action accordingly.
11. System announces voice feedback: "Navigating to coffee shop"

Alternate Flow 1 - Multi-Step Command that is difficult to follow:

1. At step 8: User enters command with multi-parts: "Find the nearest pharmacy and display me the information.
2. The system separates out command into sub-tasks.
3. System carries out pharmacy search.

4. Pharmacy information (hours, phone, etc) is retrieved by system.
5. Information to user is read in system.

Alternate Flow 2 - Unidentified Command:

1. At step 7: Command does not fit any pattern.
2. System responds: I was not in a position to comprehend. Please try again or say 'help'"
3. System is reverted to listening.
4. User has ability to repeat or seek the help of a command.

Flow 3 - Turnover Emergency Command:

1. At step 8: Command: has emergency keywords (help, SOS, emergency)
2. Emergency protocol triggered in system.
3. System by-passes processing and causes UC-11.

Postconditions:

1. There has been voice command which has been processed and carried out.
2. System has given aural feedback.
3. Action requested takes place (navigation, search, control, etc.)
4. Command history Command history logs command.

Satisfied Functional Requirements are FR6.1, FR6.2, FR6.3, FR6.4, FR6.6, NFR1.3.

UC-05: Spatial audio guidance on command.

Use Case ID: UC-06

Name of use Case: Enabling Spatial Audio Guidance.

Main Actor: Visually Impaired User.

Secondary Actors: Spatial Audio Service, Audio Hardware: Secondary Actors.

Preconditions:

1. User is in object detect or in navigation mode.
2. Permission to allow audio output.
3. User is now linked into audio output (speaker or headphones)
4. Device is supported by spatial audio hardware.

Main Flow:

1. User goes to the Settings or uses spatial audio when navigating.
2. Audio capability is done through system checks.
3. System starts spatial audio system.
4. System sets audio parameters of the 3D audio (position of listeners, orientation).
5. In the process of navigation, system identifies oncoming turn.
6. System calculates direction and distance overturn.
7. System develops spatial audio signal at relative location.
8. In case of left turn: Sound sent to the left side.
9. In case of right turn: sound on the right channel.
10. In the case of straight turn: Audio centered.
11. Directional tone in system is calculated and played at distance.
12. Audio adjustment with movement of the user:

    - Volume decreases with proximity to turn point by the user.

    - Pan size of audio is updated with heading of user.

13. On user reach of a turn point System moves to new instruction.
14. System has a sustained audio feedback on space on-course.

Alteration flow 1- Obstacle Avoidance Auditory Cue:

1. When navigating: Obstacle is detected by the system through the object identification.
2. System forms space audio warning at barrier point.
3. Direction of obstacle is indicated by audio position.
4. User can avoid barrier depending on the position of audio cue.

Alternates used flow 2 - Customer-defined sound profiles:

1. At step 1: User sets preference of audio profile.
2. System loads sound assignments of user.
3. The various types of objects or the various commands trigger various audio signatures.
4. Continue from step 5

Postconditions:

1. User is given audio feedback directionally.
2. Spatial sound is coordinated with movement of the user.
3. Audio cues alone can guide user through the user interface.
4. Sound is dynamic depending on the situation.

Functional Requirements Satisfied are FR5.1, FR5.2, FR5.3, FR5.4, FR5.5, NFR9.1, NFR9.2, NFR9.3.

UC-06: Activate Emergency Alert

Use Case ID: UC-07

Name of Use Case: Activate Emergency Alert.

Primary Actor: Visual Impaired User.

Secondary Actors: SMS Service, Emergency Service, Emergency Contacts, Location Service.

Preconditions:

1. Emergency situation (fall, lost, threatened) of the user.
2. Contacts with the emergency are established (UC-02)
3. Has cellular or internet connectivity.
4. GPS is enabled

Main Flow:

1. User identifies emergency case.
2. Emergency activation of user:
3. - Alternative A: Press and hold home button in 3 seconds.
4. - Option B: Say "Emergency" or "SOS"
5. - Alternative C: Voice command, either; Call 911/Emergency Alert.
6. System is aware of emergency trigger.
7. System validation of urgency: Emergency alert activated. Help is on the way."
8. System activates vibration pattern.
9. System gets recent GPS position.
10. System sends an emergency message with time and place.
11. System sends emergency alert to all emergency contacts through SMS.
12. Message contains: EMERGENCY: help is required by the user. Location: [GPS coords]. Time: [timestamp]"
13. System gives the user a choice:

    - 911 - Dialing straight into the emergency services.

    - "Share Location" -Send location through messaging

    - "Cancel" - disable alert in case of false alarm.

14. If user selects "Call 911":

    - System places a phone call to 911.

    - System sends out caller information to dispatcher where possible.

15. Location tracking by the system remains on in case of emergency.
16. Emergency contacts get SMS and linking to location.

17. User is given confirmation: "Emergency services notified"

Alternate Flow 1 - Fall Detection Triggers Emergency:

1. At step 1: System notifies the fall through accelerator (falls detection algorithm)
2. Emergency protocol is activated in system.
3. System alert: Fall incident. Sending emergency alert in 10sec
4. User gets 10s to cancel in case of false alarm.
5. Failure to cancel, proceed with step 3.

Alternate Flow 2 - Cellular Connection Non-existent:

1. At step 8: No connection to cellular device.
2. System tries to transmit through internet (SMS over IP)
3. In case of internet, forward to friends through email/ application notification.
4. Continue with other steps

False Emergency: Accepted Flow 3 - Cancel:

1. At step 10: User selects "Cancel"
2. Sends Emergency alert before cancelling system.
3. Display in the system: Emergency alert cancelled.
4. Incident to be analyzed in system logs.

Postconditions:

1. Contacts with emergency authorities are provided.
2. It can involve the emergency services (911).
3. Location tracking of the user is being done.
4. Emergency mode would stay on till resolved by a user or emergency services.
5. To keep log records, an incident is recorded at time and place.

FR7.1, FR7.2, FR7.3, FR7.4, FR7.5, FR7.6, NFR4.3, NFR4.4 Satisfied.

UC-07: Save and Recall Route

Use Case ID: UC-08

1. Save and Recall Route Use Case Name: Save and Recall Route.
2. Primary Actor: Visually impaired User.
3. Jacks of Stations: Local Storage, Route Memory Service.

Preconditions:

1. User has finished a navigation session (destined or has stopped navigation)
2. User has covered long distance (100 meters or more).
3. Memory contains route data of the system.

Main Flow:

1. Navigating to destination is completed by user.
2. System proclaims: You have reached the end of the road
3. System provides a display of route summary (distance, time, turns).
4. istiqsaan requests: Save this route: <|human|>user requests: Save this route: Save this route or voice request Remember this path.
5. System prompts of route name: What would you like to name this route?
6. Name: route to Coffee Shop or Home to Office User gives name of the route: Route to Coffee Shop or Home to Office
7. System confirm: Saving route as [name]
8. System stores store system route data on a local basis:

    - Start point, end point

    - All waypoints and turns

    - Hazards in route identified.

    - Time of day recorded

9. Confirmation of a successful save of a route is given by the system: " Route saved successfully"
10. Subsequently, a request by a user recalls: "Go to previously saved route Coffe Shop.
11. System retrieves routing in store.
12. System confirms that the current position is close to the start of the route.
13. System commences guided navigation with stashed route.
14. System offers common direction according to the route history.

Alternate Flow 1 - Modify Saved Route:

1. At step 10: User clicks on Manage Routes.
2. List of saved routes is shown on the system.
3. User selects route to edit
4. System permits route name, start end point to be changed.
5. User confirms changes
6. System updates route

Alternate Flow 2 - Delete Route:

1. At step 10: User clicks on the option of Manage Routes.

2. System consists of list of saved routes.
3. User selects "Delete route"
4. System confirms deletion
5. User confirms
6. Route gets removed out of the local storage.

Alternate Flow 3 - Route Not Available (Offline):

1. On step 11: There is no internet connection on device.
2. System uses route local cache data.
3. Nevertheless, in case route is in place locally, it is followed.
4. Otherwise, system downloads on user request.

Postconditions:

1. Route stored locally to be used in the future.
2. Direction is easily remembered without having to search again.
3. User has swifter navigation on commonly used routes.
4. System has acquired general user paths in intelligent suggestions.

Meetings of the functional requirements FR12.1, FR12.2, FR12.3, FR9.1, FR9.2.

UC-08: Read Text with OCR

Use Case ID: UC-09

Name of Use Case: Read with OCR Text.

*Primary Actor :*Visually Impaired User.

Secondary Actors: Camera, OCR Service and Text-to-speech Service.

Preconditions:

1. Permission to access the camera has been granted to the user.
2. The user is in the place where there is an observable text (sign, document, label).
3. Device has camera access
4. OCR model is loaded

Main Flow:

1. User goes to AR Screen or pronounces "Read text" or "Scan".

2. System activates camera
3. System shows real time camera feed.
4. User makes an orientation to text.
5. Video frames are constantly being processed by the system.
6. On detecting text with confidence of above 80 percent:

   - Texts in AR view are identified by system.

   - System identifies text through text-to-speech.

7. Text options are available in the system:

   -Translating- Translating to other language

   - "Copy" - Copy text to clipboard

   - "Search" - On-line search of text.

   - "More" - Additional options

8. User makes a choice through voice command.
9. In case of there is translation: System translates the text and pronounces it aloud.
10. When: Search: System searching Found text and read results.
11. User is able to move about to other text on the same screen.

Alternate Flow 1 - Document Scanning:

1. At step 2: User clicks on Scan Document.
2. System master switches to camera in document mode.
3. User can press to save page.
4. System does OCR on entire page.
5. System reads the content of the whole page.

Alternate Flow 2 -Continuous OCR Mode:

1. In step 1: User switches on the Continuous Reading.
2. System reads anything it comes across.
3. System offers real time text feedback.

Postconditions:

1. Text has been read to user
2. User can access translation or other information.
3. A text data is able to be saved to be referred to in the future.

FR11.1, FR11.2, FR11.3, FR11.4 Satisfied.

UC-9: Connect Wearable Device

Use Case ID: UC-10

Name of Use Case: Connect Wearable device.

Key Personality: Visually Impaired User.

*Secondary Actors :*Bluetooth Service, Wearable Device.

Preconditions:

1. Wearable device is on and on pairing mode.
2. Smartphone has Bluetooth activity.
3. The range of wearable device is less than Bluetooth (~10 meters).
4. User is enabled with Bluetooth.

Main Flow:

1. User goes to Settings Screen.
2. User chooses the Bluetooth or Connected Devices.
3. System will show a list of available devices via Bluetooth.
4. Bluetooth scan commences in system.
5. Wearable device (e.g., "SENSEI Haptic Band") is found in the system.
6. System reads and identifies devices to user.
7. User voice-selects device: 1- Connect to Haptic Band or 1- First device.
8. Paring request is initiated to wearable by the system.
9. Wearable device notifies user to confirm that it is paired (this may need a button press).
10. Wearable device user confirmation.
11. System provides secure Bluetooth linkage.
12. Credential-to-credential system exchanges.
13. System responses:.... connected to [device name] successfully.
14. System tests interrelates with haptic feedback test pattern.
15. User experience haptic pulse that is connected.
16. System also has device pairing as auto-connect in the future.

Alternate Flow 1 - Entry of the devices manually:

1. In step 3: User selects Pair New Device.
2. System goes into manual pairing.
3. User enters or types in device address or name.
4. System tries to connect to given device.

Connection Failure on Alternate Flow 2:

1. At step 11: Connection fails
2. System announcements: Connection failure. Please turn device on and keep it close at hand please.
3. System gives troubleshooting measures.
4. User is allowed to retries or to choose another device.

Postconditions:

1. Wearable device is associated and connected.

2. System is able to provide haptic feedback to wearable.
3. Connection is maintained between future app sessions.
4. Wearable can be used to improve navigation haptic-feedback.

The functional requirements satisfied include FR10.1, FR10.2, FR10.3, and FR10.4.

UC-10: Adjust Audio and Accessibility learning disabilities.

Use Case ID: UC-12

Name of Use Case :Customize Audio and Accessibility Settings.

Key Personality: Visually Impaired User.

Secondary Actors : Audio System, Secondary Actors.

Preconditions:

Logged in and on Settings Screen.

User has found his/her way to Accessibility or Audio settings.

Main Flow:

1. User opens Settings Screen
2. System includes voice menu to show settings categories.
3. User picks either Audio settings or accessibility.
4. Audio settings which are shown in the system:

   - Speech speed (0.5x to 2.0x)

   - Speech pitch (adjustable)

   - Audio volume (0-100%)

   - Speaker (speaker, headphones, headphones)

5. User can change the pace of speech: "Change speech to 1.5x"
6. System confirms: "Speech speed=1.5 x"
7. System preview applies setting.
8. 99 different user tests environment: "Read test message"
9. Test message with new settings System plays.
10. Further confirmation or adjustment is done by the user.
11. User changes volume: "Volume 80% on it"
12. System confirms and applies
13. User opens up to other settings:
14. - Haptic feedback On/Off

    - Audio cues style

    - Emergency alert volume

    - Level of verbosity of screen readers.

15. User saves all settings
16. System affirmation: "Savings done successfully"
17. Future sessions have settings that are maintained.

Alternate Flow 1 -Reset to Default:

1. At step 14: User commands "Reset to defaults"
2. System confirms action
3. Individual settings are restored to default settings.
4. User can confirm either reset or cancel.

Flow 2 - Profile of import settings:

1. On step 1: User clicks on Import Settings.
2. Preset profiles are shown on the system screen.
3. User chooses profile (e.g., "Quiet Environment, Noisy Commute" etc.).
4. Imports system and all settings of profile.
5. Still can be customized by user.

Postconditions:

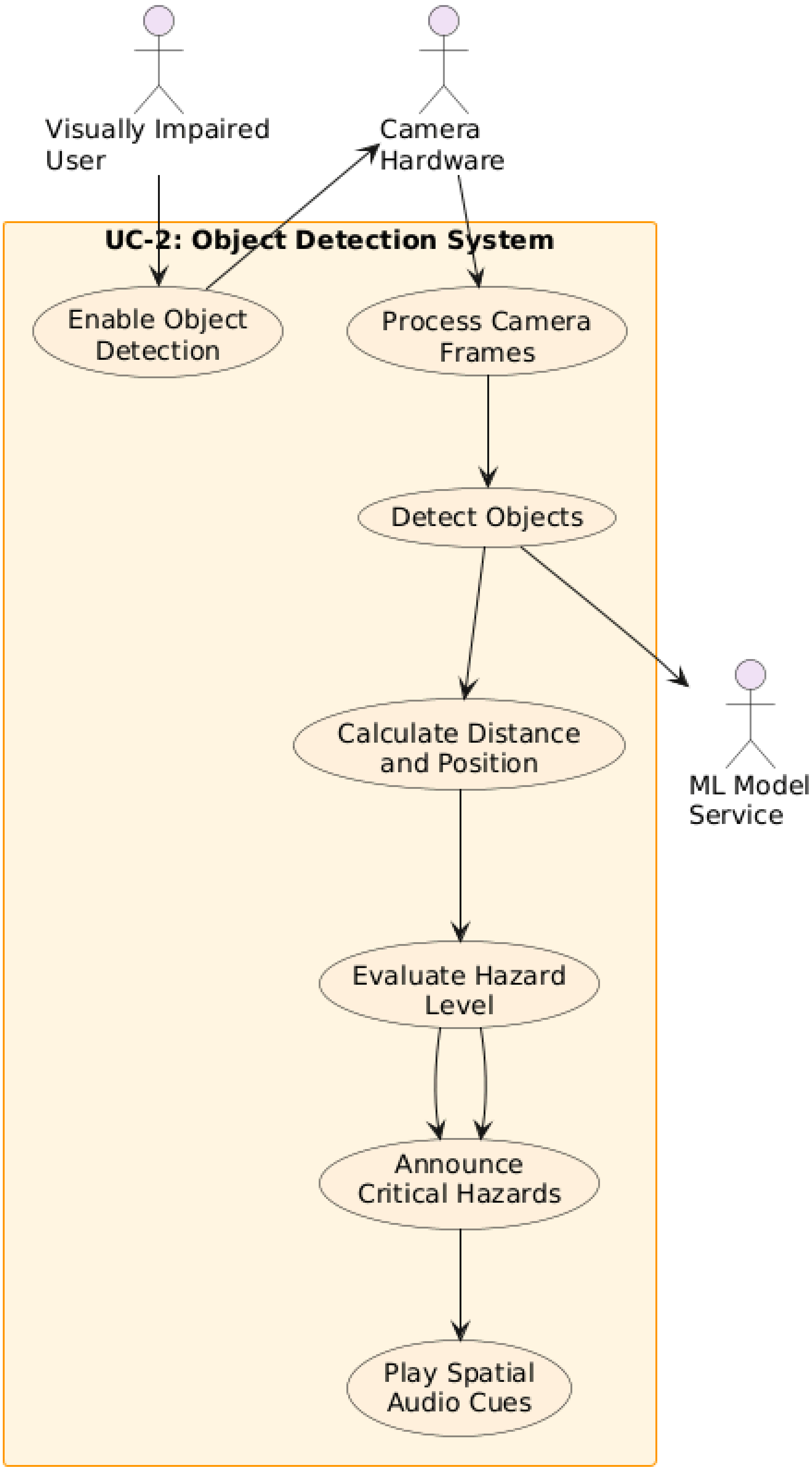
1. Audio preferences of user are stored.
2. The sound of all apps is made with tuned settings.
3. The user has an ideal accessibility set-up.
4. The settings are maintained between the app activities and device reboots.

FR1.4, FR2.1, FR5.1, FR6.6, NFR2.3, NFR2.4, NFR9.1 Satisfied.

System Design

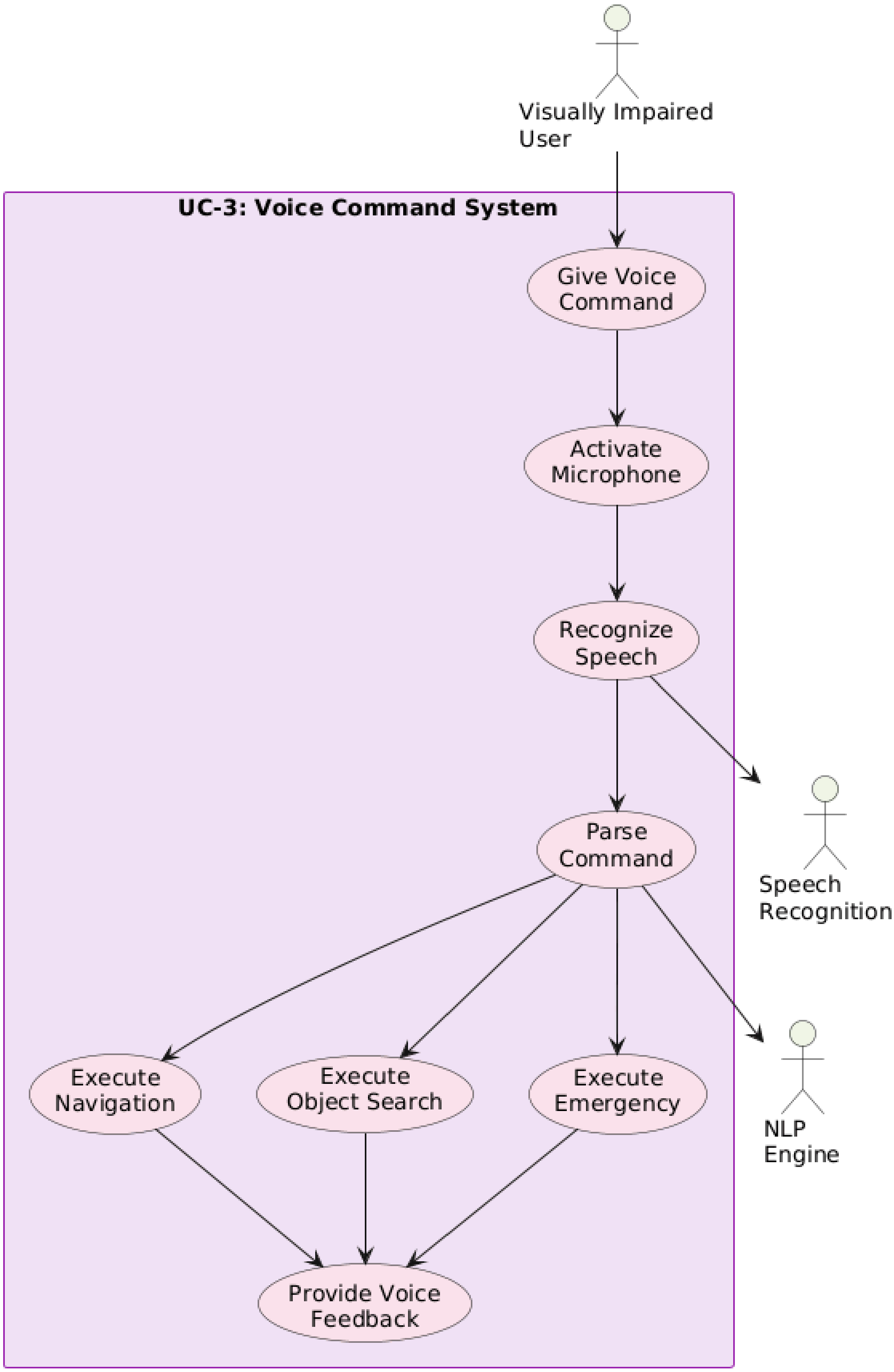
1.3 Use Case Design

1.3.1 UC-1:



This figure shows the detection pipeline of objects. In the case where the user switches on object detection, the system takes frames on the camera, and uses a machine learning model to find the objects, compares their distance and position, and assesses the degree of hazards and alerts users about the high-risk zones through spatial audio loops.

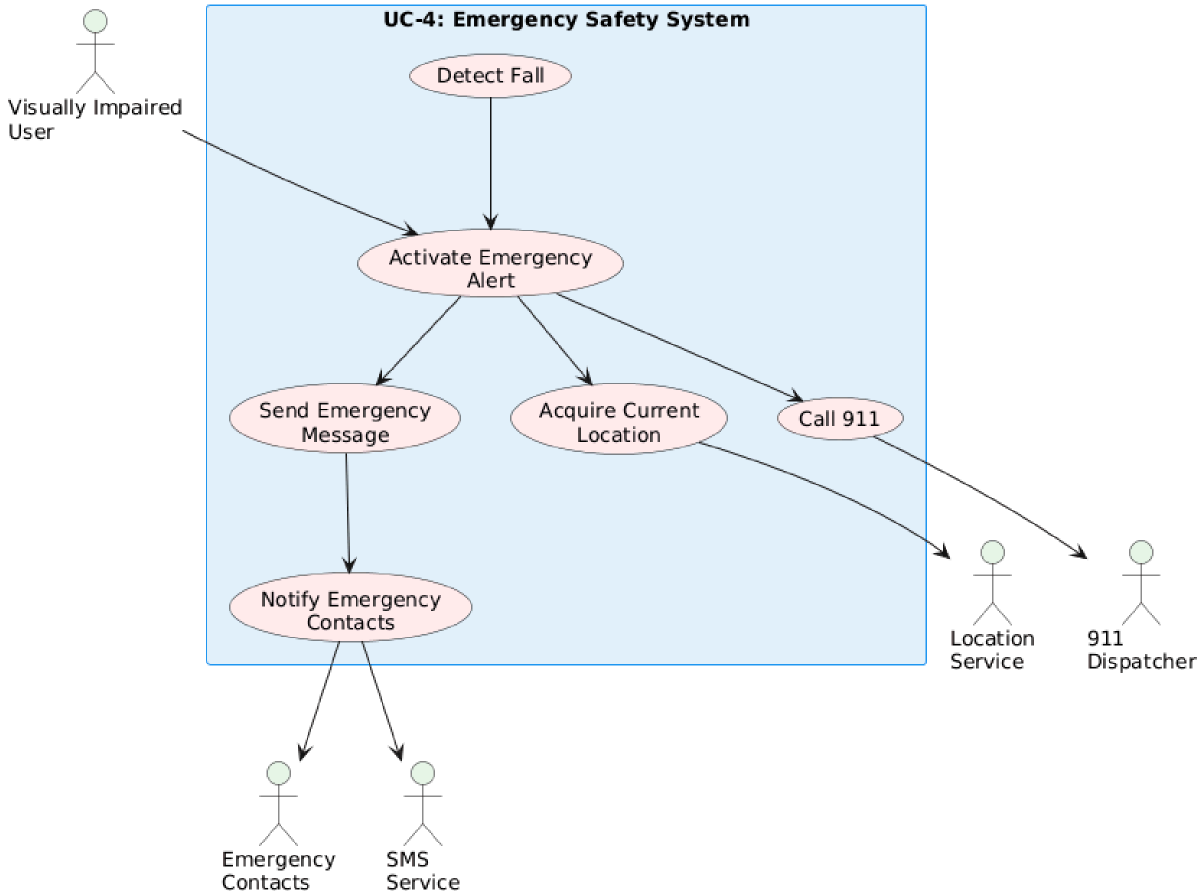
1.3.2 UC-2:



Visually Impaired User
UC-3: Voice Command System
Give Voice Command
Activate Microphone
Recognize Speech
Speech Recognition
Parse Command
NLP Engine
Execute Navigation
Execute Object Search
Execute Emergency
Provide Voice Feedback

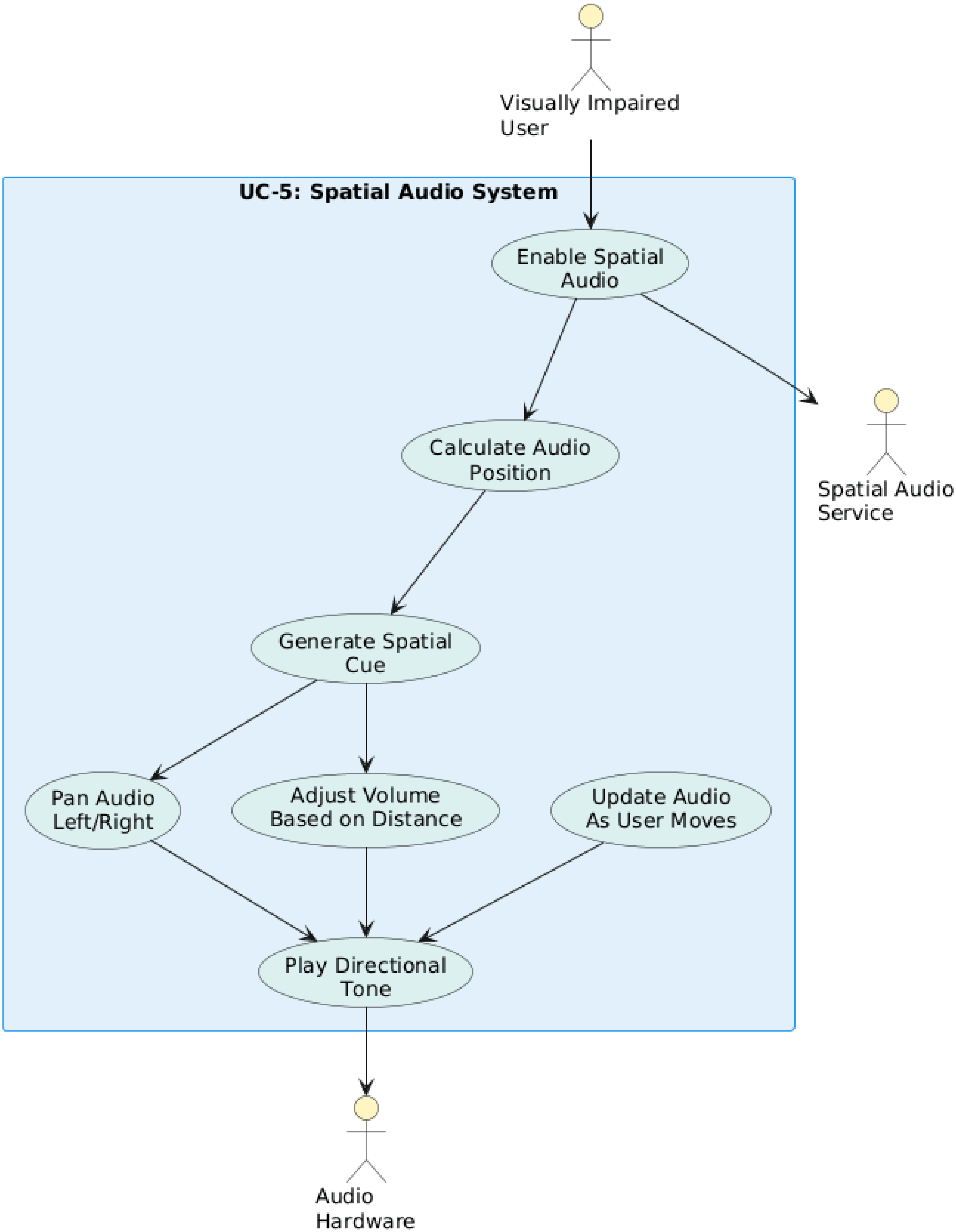
This diagram illustrates the voice command processing. A voice command is issued by the user and the microphone is turned on and the speech is recognized and translated into text, the NLP engine breaks the command into the action type and the action (navigation, search, emergency) is performed. The system then gives voice confirmation regarding the action.

1.3.3 UC-3:



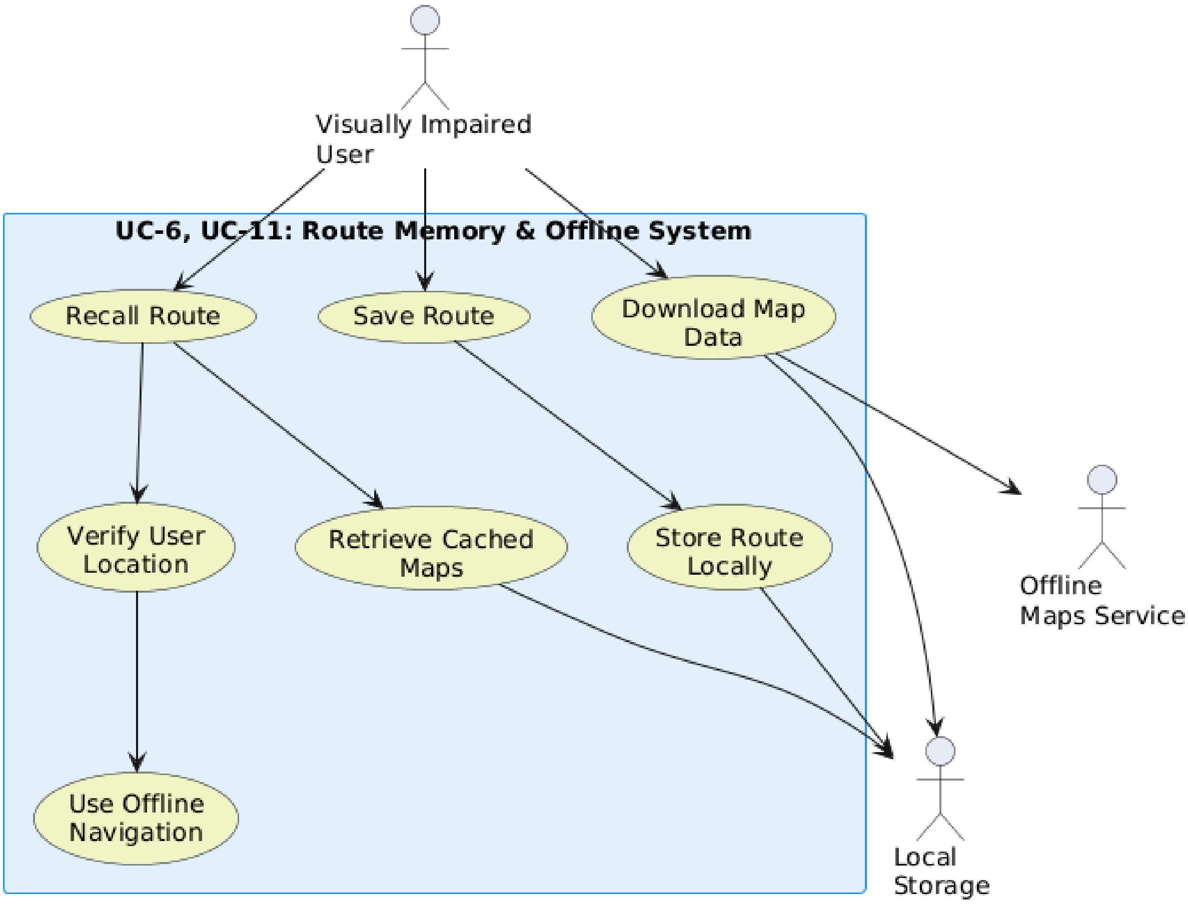
This chart depicts Emergency alert processes. An emergency alert can be activated manually by the user or can be detected automatically by the system in case of a fall. Once activated, the system will get the user location, send emergency messages to contacts and be able to call 911. Every emergency contact is informed through SMS.

1.3.4 UC-4:



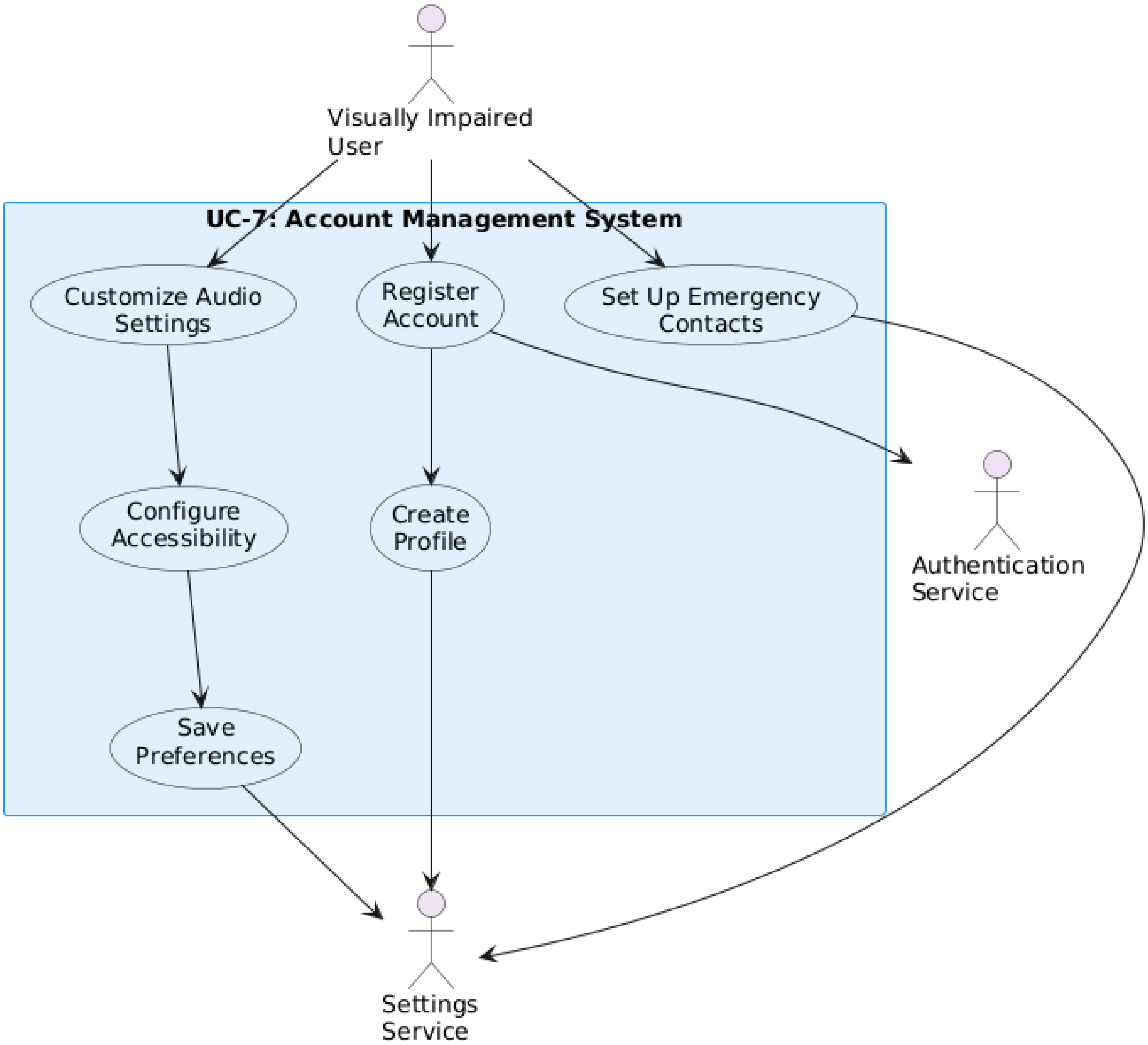
The following diagram illustrates the operation of spatial audio system. When you turn it on, the system will determine where each sound will be produced, it could be in the left, right, centre, front or behind you. It then constructs the appropriate spatial cues, directs the sound to the appropriate channel and modulates the volume according to a distance of a target. The sound remains real time updated as you move around thus everything seems to be naturally related to where you are.

1.3.5 UC-5:



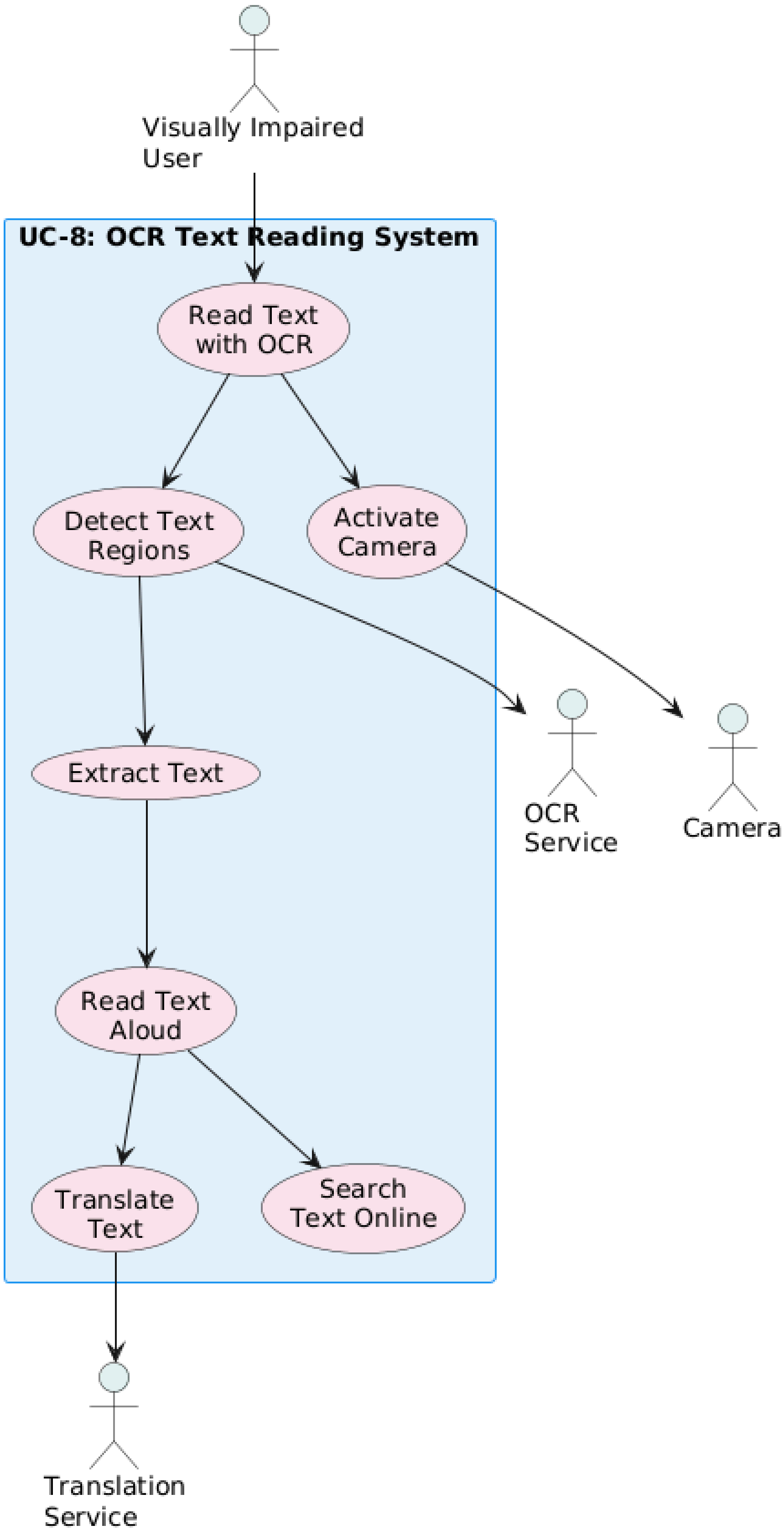
This diagram is a step by step explanation of the way route memory and offline navigation operate. You have the option of saving routes on your device and they are there till the next time you require them. On returning to a route that has been saved, the system has an ability to retrieve it and verify that it is still correct. It also allows you to download map data so that you can use it when offline, thus, being able to maneuver even when you are without the internet connection.

1.3.6 UC-6:



This flow chart takes one through the account management and establishment flow. Once an individual registers, the system will generate his/her account and liaise with the authentication service to prepare it all. Then, they have an opportunity to include emergency contacts and to change their accessibility preferences, such as audio preferences. All such decisions are saved in the settings service to be able to be used again when the user comes in.

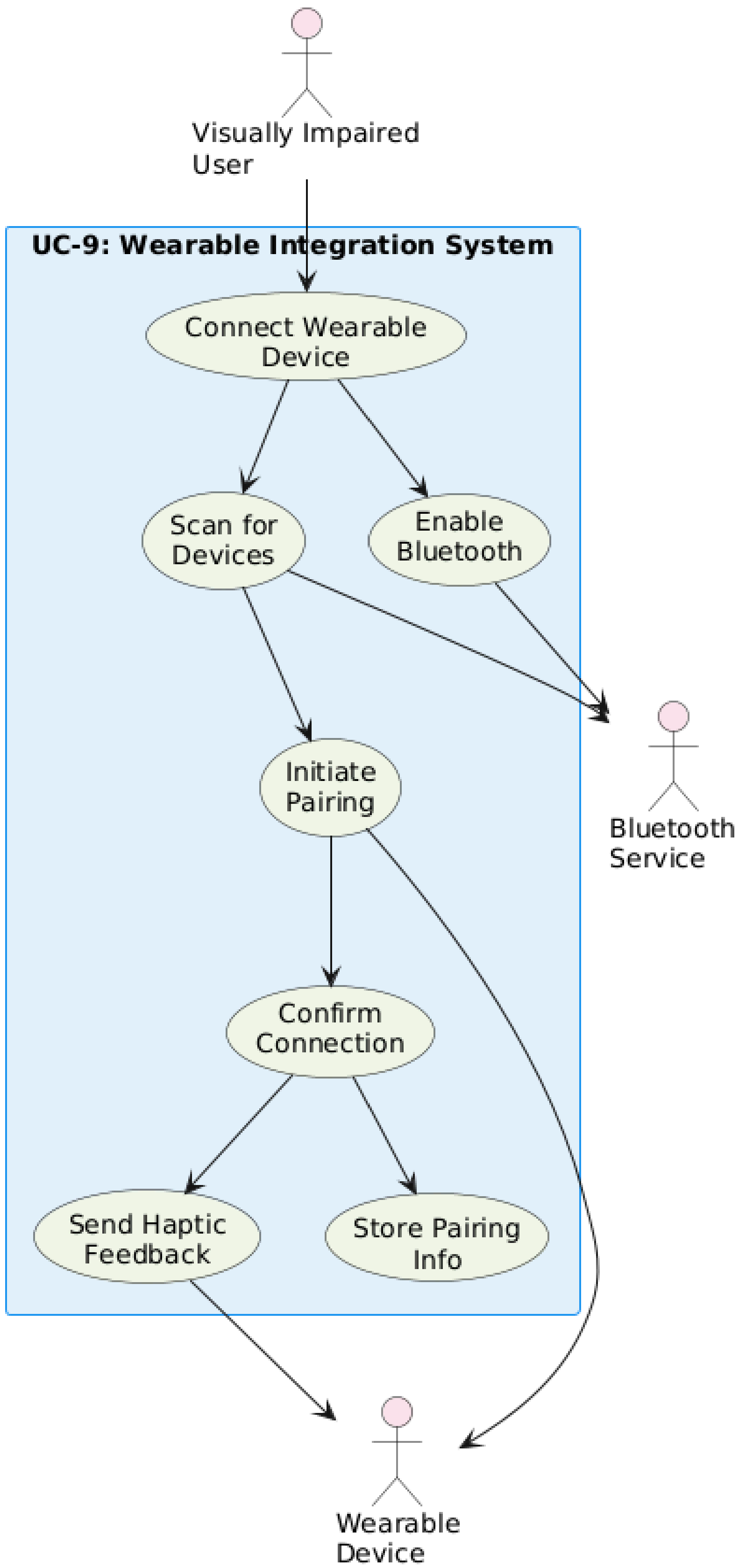
1.3.7 UC-7:



This is a diagram of the way in which the OCR workflow is assembled. When an individual directs the system to read the text, the camera switches on and searches any places where there is writing.

When such areas have been located, the OCR service extracts the text and reads it aloud. There, a user can request a translation as well as an online search depending on the identified text.

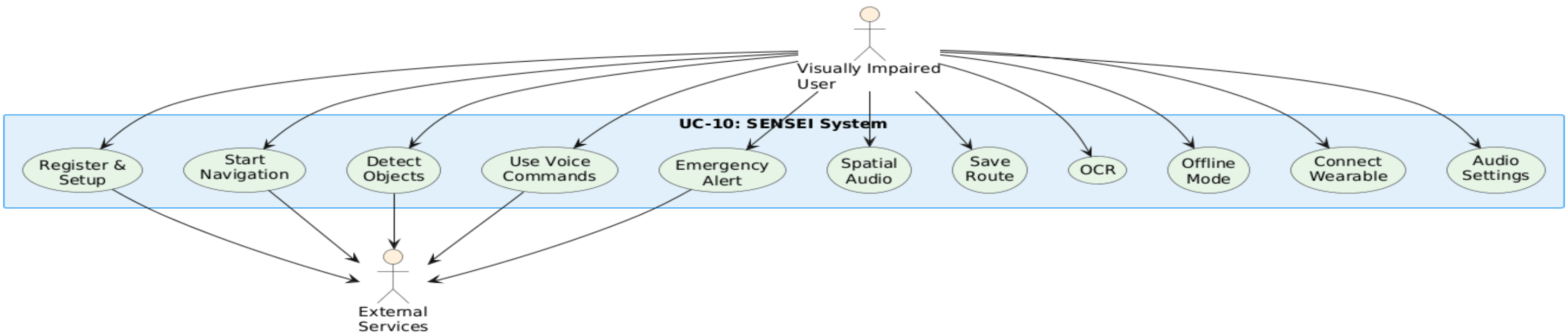
UC-8:



The following diagram guides the way the wearable device links. The user switches on the Bluetooth and the system begins to scan until it detects the wearable that is nearby. It then proceeds to pair the two and stores the pairing information such that it can be used in the future to connect once again.

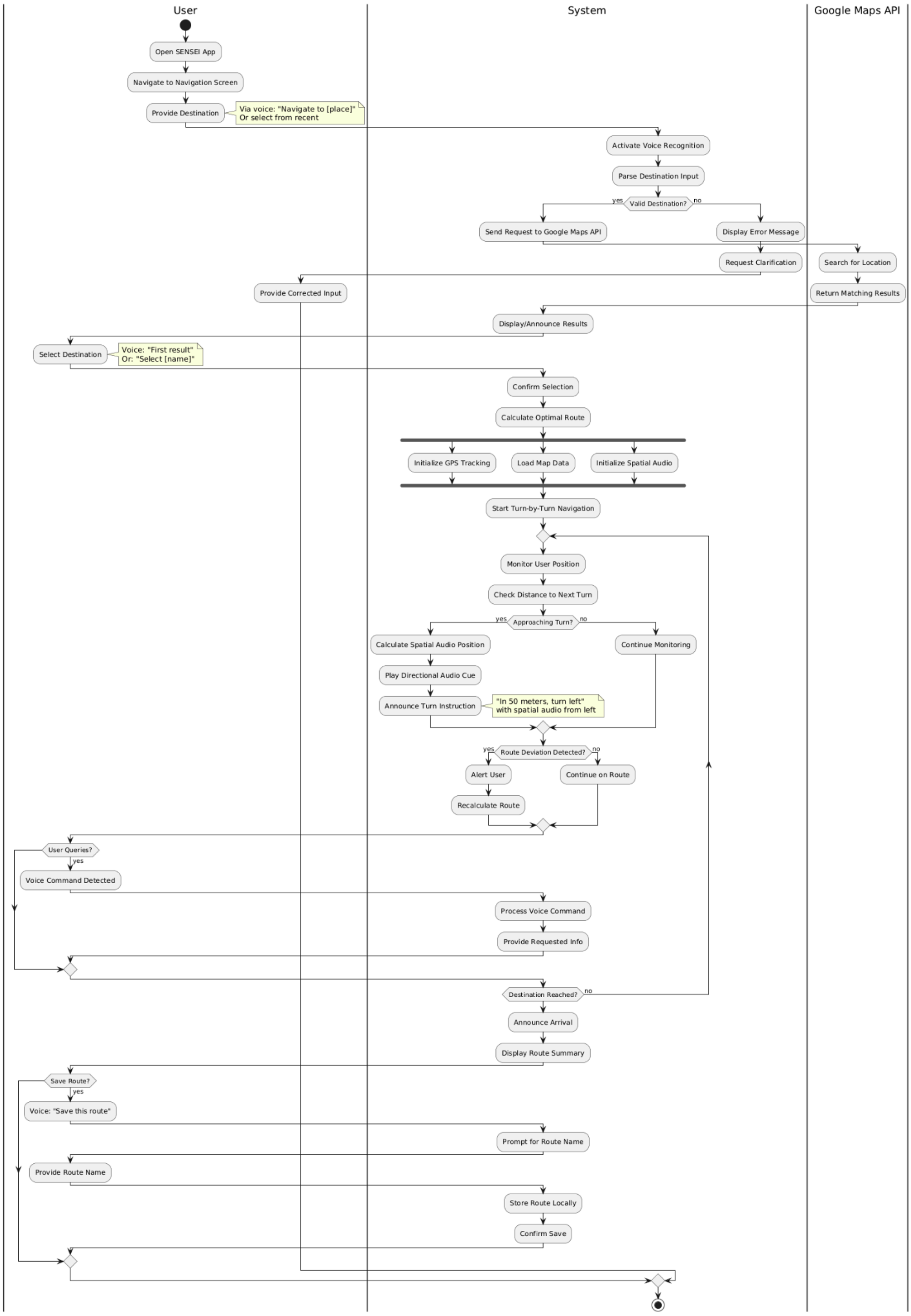
When all the connections are made the system will be in a position to provide the haptic feedback directly to the wearable.

1.3.8 UC-9:



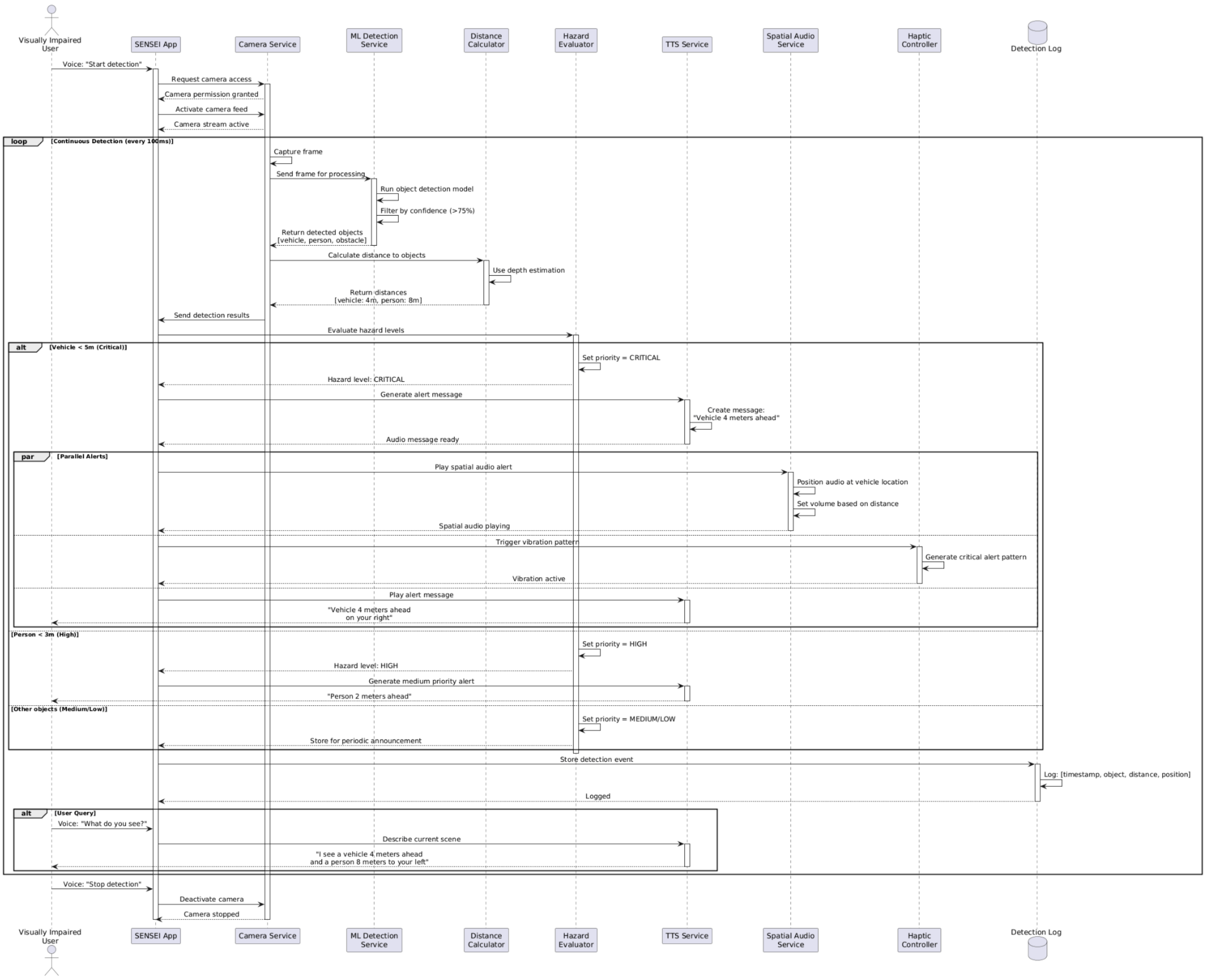
This diagram brings together all the key use cases and indicates the flow of the user between the various sections of the system. It provides a clear understanding of the extent to which the system can achieve in addition to indicating the activities that depend on external services.

1.4 Activity Diagram



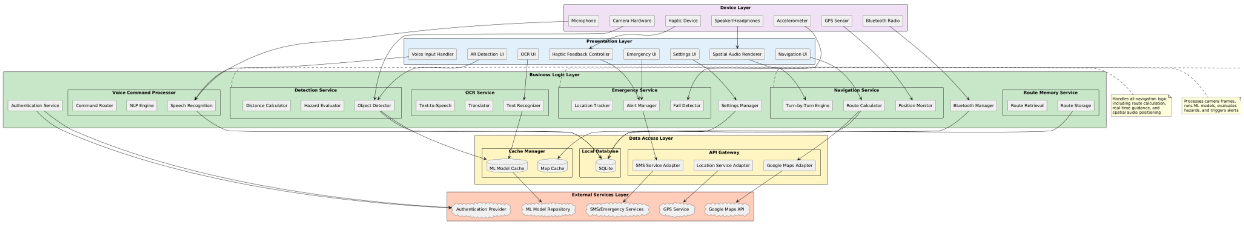
This diagram represents the entire process of navigation, which begins at the point when the user types in a destination. It includes the route calculation, constant position tracking, and turn-by-turn guidance with the help of spatial audio. It has also optional features such as saving a route. There are parallel activities occurring simultaneously like GPS tracking and map data loading, as well as decision points, which verify such things as whether the user is close to a turn or has out of the road, or if the user posed a question etc.

1.5 Sequence Diagram



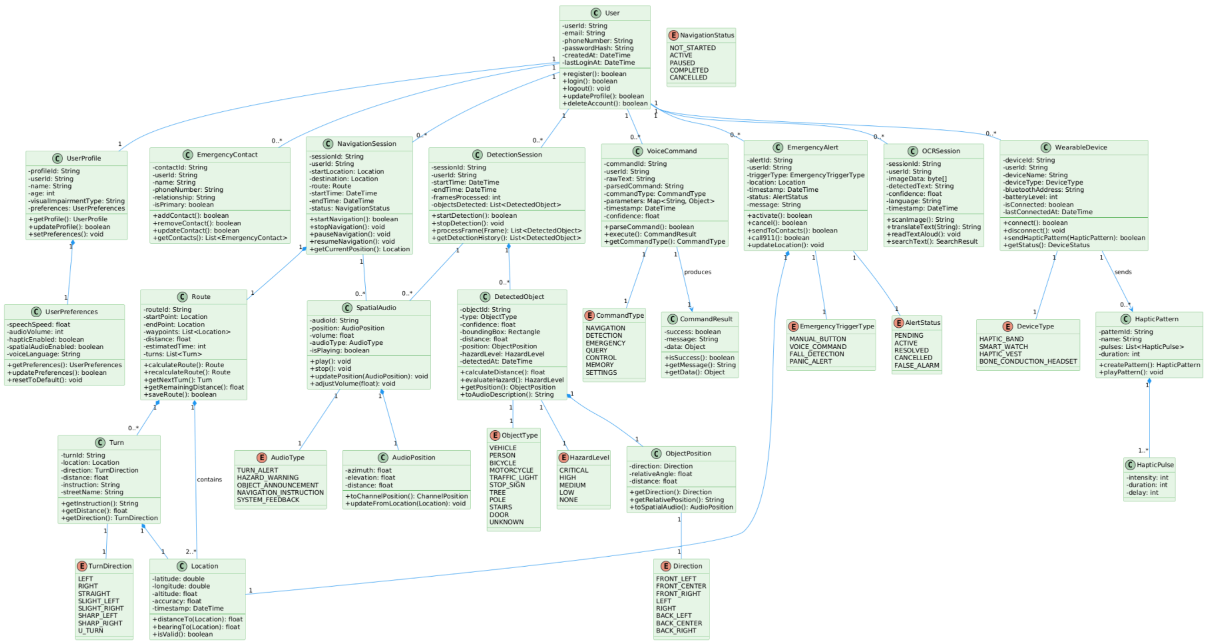
This diagram takes one through the process of how the system identifies objects, and invokes hazards. It illustrates the continuous cycle between processing camera frames, camera objects are identified by the machine learning model, they are detected, system measures the distance between the objects, and, based on the distance, they are determined to be hazardous or not. The user receives notifications in the form of sound, spatial or haptic in response to the severity of the risk involved.

1.6 Software Architecture Diagram



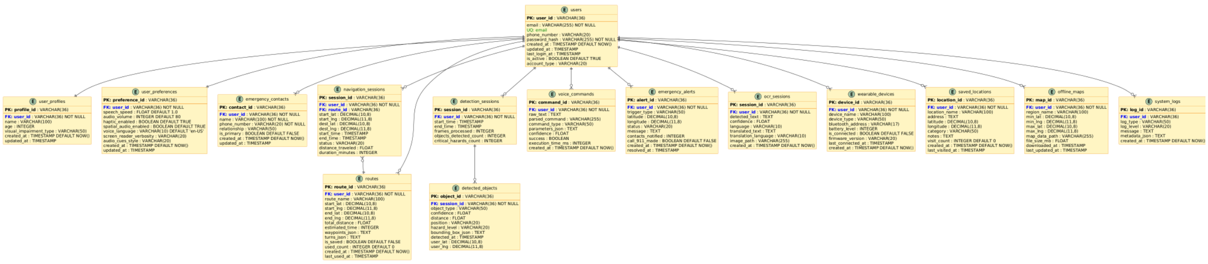
The following diagram illustrates the construction of the SENSEI system as layers. The top layer is the presentation, which deals with user interface and dealings with the user. The business logic layer is found below that, and contains the primary services and system rules. The second layer oversees data and communication with the database and APIs. It also has a layer of external services and the device layer is the real hardware that the system is based on. The design is such that it has its own job in each layer with the flow proceeding down to the user side to the outside services.

1.7 Class Diagram



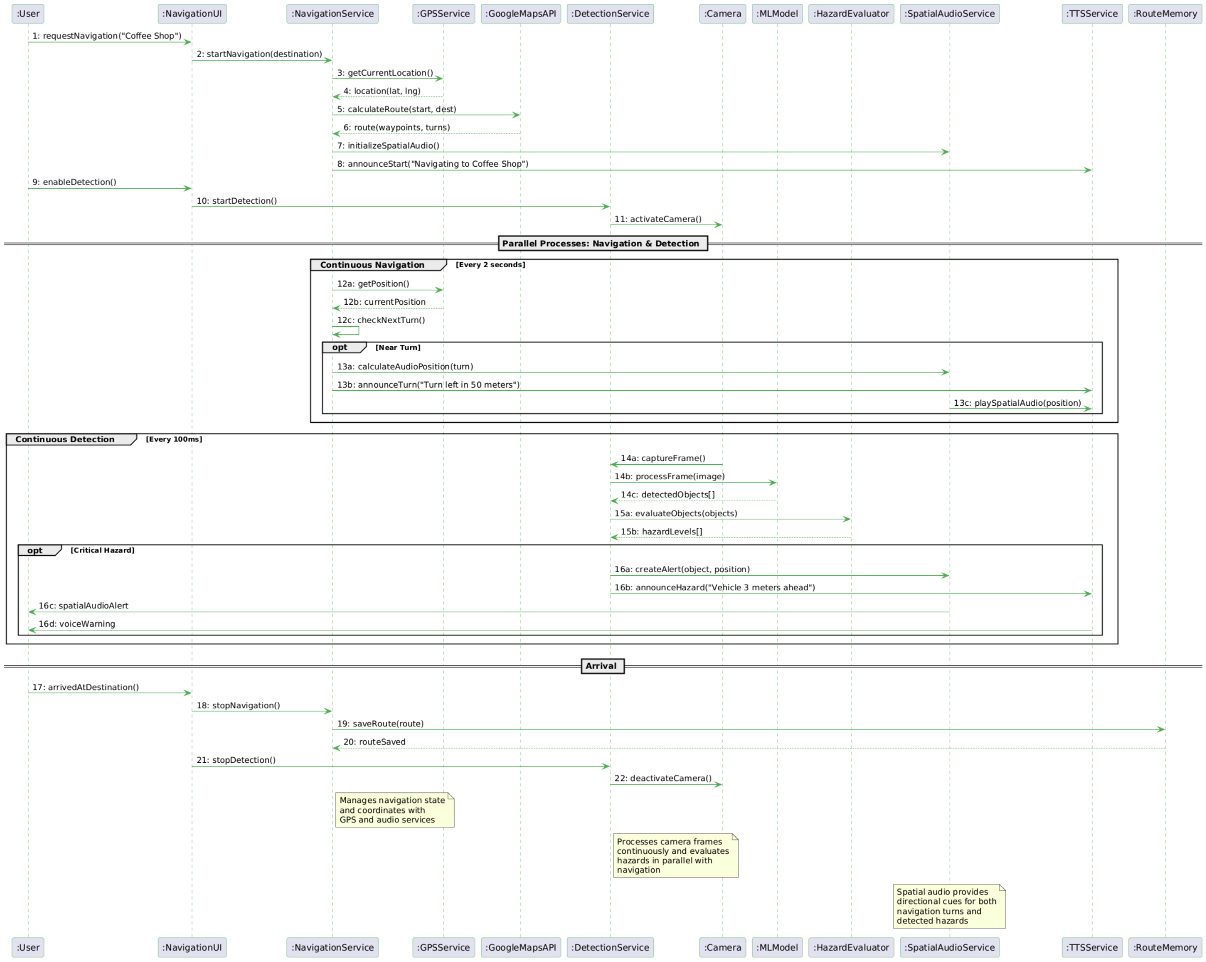
This illustration displays the key classes to constitute the SENSEI system based on their usage i.e. user management, navigation, object detection, voice commands, spatial audio, emergency feature, OCR and wearable devices. All the important data and methods are listed in a class, and the lines between them demonstrate the connections and mutual dependencies between them.

1.8 Database Diagram



The following diagram illustrates the key tables within the SENSEI system and the contents of each, including the columns, types of data, and the relationships between the tables. It is designed to operate under all the requirements of the system such as user management, navigation tracking, object detection, voice commands, emergency notification, text reading using OCR, wearable integration and functionality even when the device is not online.

1.9 Collaboration Diagram



This diagram illustrates the communication of the various components of the system with each other as the system navigates as well as the object detection. It follows the process of every process one after the other, demonstrating that the Navigation Service and Detection Service perform concurrently. It is possible to observe how GPS tracking, route planning, object detection camera, hazard checks, and spatial audio functions can all cooperate and help the user feel safe and clear.

System Testing

1.10 Test Cases Design

1.1 Testing Objectives

The primary objectives of SENSEI system testing are:

1. **Functional Correctness**: Verify all features work as specified in requirements
2. **Accessibility Validation**: Ensure the app is fully accessible for visually impaired users
3. **Performance Verification**: Validate response times and real-time processing capabilities
4. **Safety Assurance**: Confirm emergency features function reliably
5. **Usability Testing**: Ensure intuitive voice-based interaction
6. **Reliability Testing**: Validate system stability under various conditions

1.2 Test Case Structure

Each test case includes:

Field	Description
Test ID	Unique identifier
Test Name	Descriptive name
Module/Feature	Component being tested
Priority	Critical/High/Medium/Low
Type	Unit/Integration/Acceptance
Preconditions	Required setup
Test Steps	Detailed execution steps
Test Data	Input data required
Expected Result	Expected outcome
Actual Result	To be filled during execution
Status	Pass/Fail/Blocked
Requirements Coverage	FR/NFR references

1.11 Unit Testing

2.1 User Management Module

UT-USER-001: User Registration with Valid Email

Field	Details
Test ID	UT-USER-001
Test Name	Register User with Valid Email
Module	User Management
Priority	Critical
Preconditions	- App installed - No existing account with test email
Test Steps	1. Launch app 2. Navigate to registration screen 3. Enter valid email: test@sensei.com 4. Enter password: SecurePass123! 5. Confirm password 6. Submit registration
Test Data	Email: test@sensei.com Password: SecurePass123!
Expected Result	- User account created successfully - Confirmation message displayed - User redirected to home screen - User logged in automatically

UT-USER-002: User Registration with Invalid Email

Field	Details
Test ID	UT-USER-002
Test Name	Register User with Invalid Email Format
Module	User Management
Priority	High
Preconditions	- App installed
Test Steps	1. Launch app 2. Navigate to registration screen 3. Enter invalid email: invalid-email 4. Enter password: SecurePass123! 5. Submit registration
Test Data	Email: invalid-email

	Password: SecurePass123!
Expected Result	- Registration fails - Error message: "Invalid email format" - Email field highlighted - User remains on registration screen

UT-USER-003: Password Strength Validation

Field	Details
Test ID	UT-USER-003
Test Name	Validate Weak Password Rejection
Module	User Management
Priority	High
Preconditions	- Registration screen open
Test Steps	1. Enter valid email 2. Enter weak password: 123 3. Attempt to submit
Test Data	Password: 123
Expected Result	- Registration blocked - Error: "Password must be at least 8 characters" - Password requirements displayed

UT-USER-004: Emergency Contact Addition

Field	Details
Test ID	UT-USER-004
Test Name	Add Emergency Contact Successfully
Module	User Management
Priority	Critical
Preconditions	- User logged in - Settings screen accessible
Test Steps	1. Navigate to Settings → Emergency Contacts 2. Click "Add Contact" 3. Enter name: "John Doe" 4. Enter phone: "+1-555-0100" 5. Save contact
Test Data	Name: "John Doe" Phone: "+1-555-0100"

Expected Result	- Contact saved successfully - Contact appears in list - Confirmation message displayed

UT-USER-005: Duplicate Emergency Contact Prevention

Field	Details
Test ID	UT-USER-005
Test Name	Prevent Duplicate Emergency Contact
Module	User Management
Priority	Medium
Preconditions	- User logged in - One emergency contact already exists: "+1-555-0100"
Test Steps	1. Navigate to Emergency Contacts 2. Click "Add Contact" 3. Enter same phone: "+1-555-0100" 4. Attempt to save
Test Data	Phone: "+1-555-0100"
Expected Result	- Save fails - Error: "Contact already exists" - User prompted to edit existing contact

2.2 Object Detection Module

UT-DET-001: Object Detection Model Initialization

Field	Details
Test ID	UT-DET-001
Test Name	Initialize ML Detection Model
Module	Detection Service
Priority	Critical
Preconditions	- App installed - ML model file exists in assets
Test Steps	1. Call initializeDetectionModel() 2. Wait for model loading 3. Check model status
Test Data	Model file: object_detection_v3.tflite

Expected Result	- Model loads successfully - Status: MODEL_READY - Model version displayed - Load time < 3 seconds

UT-DET-002: Object Detection from Image Frame

Field	Details
Test ID	UT-DET-002
Test Name	Detect Objects in Test Image
Module	Object Detector
Priority	Critical
Preconditions	- Detection model loaded - Test image available (contains car and person)
Test Steps	1. Load test image test_scene_01.jpg 2. Call detectObjects(image) 3. Parse results
Test Data	Test image with: 1 vehicle, 2 persons
Expected Result	- Detects 1 vehicle (confidence > 75%) - Detects 2 persons (confidence > 75%) - Returns bounding boxes for each - Processing time < 200ms

UT-DET-003: Distance Calculation to Detected Object

Field	Details
Test ID	UT-DET-003
Test Name	Calculate Distance to Object
Module	Distance Calculator
Priority	Critical
Preconditions	- Camera calibrated - Object detected with bounding box
Test Steps	1. Pass detected object with known size 2. Call calculateDistance(object) 3. Compare with ground truth
Test Data	Known distance: 5 meters
Expected Result	- Calculated distance: 4.5-5.5 meters - Error margin < 10% - Calculation time < 50ms

UT-DET-004: Hazard Level Evaluation - Critical

Field	Details
Test ID	UT-DET-004
Test Name	Evaluate Critical Hazard Level
Module	Hazard Evaluator
Priority	Critical
Preconditions	- Object detected: Vehicle - Distance calculated: 3 meters
Test Steps	1. Create DetectedObject (Vehicle, 3m) 2. Call evaluateHazard(object) 3. Check hazard level
Test Data	Type: VEHICLE, Distance: 3 meters
Expected Result	- Returns HazardLevel.CRITICAL - Alert priority set to highest - Immediate notification triggered

UT-DET-005: Hazard Level Evaluation - Low

Field	Details
Test ID	UT-DET-005
Test Name	Evaluate Low Hazard Level
Module	Hazard Evaluator
Priority	Medium
Preconditions	- Object detected: Tree - Distance: 15 meters
Test Steps	1. Create DetectedObject (Tree, 15m) 2. Call evaluateHazard(object) 3. Check hazard level
Test Data	Type: TREE, Distance: 15 meters
Expected Result	- Returns HazardLevel.LOW - No immediate alert - Object logged for context

UT-DET-006: Object Position Determination

Field	Details
Test ID	UT-DET-006

Test Name	Determine Object Position Relative to User
Module	Object Position Calculator
Priority	High
Preconditions	- Object detected in frame - Bounding box coordinates available
Test Steps	1. Object at X position: 100 pixels (left third of frame) 2. Call calculatePosition(boundingBox) 3. Check position classification
Test Data	BBox center X: 100 (frame width: 640)
Expected Result	- Returns Direction.FRONT_LEFT - Relative angle calculated correctly - Position string: "on your left"

2.3 Voice Command Module

UT-VOICE-001: Voice Command Recognition - Emergency

Field	Details
Test ID	UT-VOICE-002
Test Name	Recognize Emergency Voice Command
Module	Voice Command Processor
Priority	Critical
Preconditions	- Voice recognition active
Test Steps	1. Input audio: "Emergency help" 2. Process command 3. Verify emergency trigger
Test Data	Audio: "Emergency" or "Help" or "SOS"
Expected Result	- Command type: EMERGENCY - Emergency protocol activated immediately - Priority set to CRITICAL - Confidence > 90%

UT-VOICE-002: Voice Command Confidence Threshold

Field	Details
Test ID	UT-VOICE-003
Test Name	Reject Low Confidence Voice Commands
Module	Voice Command Processor

Priority	High
Preconditions	- Voice recognition active
Test Steps	1. Input unclear/garbled audio 2. Process command 3. Check confidence score
Test Data	Garbled audio with background noise
Expected Result	- Confidence < 60% - Command rejected - User prompted: "I didn't understand, please repeat" - No action executed

UT-VOICE-003: Command Parameter Extraction

Field	Details
Test ID	UT-VOICE-004
Test Name	Extract Parameters from Voice Command
Module	NLP Engine
Priority	High
Preconditions	- NLP engine initialized
Test Steps	1. Input: "Navigate to Starbucks on Main Street" 2. Parse command 3. Extract parameters
Test Data	Command: "Navigate to Starbucks on Main Street"
Expected Result	- Action: NAVIGATE - Destination: "Starbucks" - Location detail: "Main Street" - Parameters extracted correctly

2.4 Spatial Audio Module

UT-AUDIO-001: Spatial Audio Position Calculation

Field	Details
Test ID	UT-AUDIO-001
Test Name	Calculate Spatial Audio Position
Module	Spatial Audio Service
Priority	High
Preconditions	- Spatial audio initialized - Object position known

Test Steps	1. Object at 45° to the right 2. Call calculateAudioPosition(45°, 10m) 3. Check audio positioning
Test Data	Azimuth: 45°, Distance: 10m
Expected Result	- Azimuth set to 45° - Elevation: 0° - Volume calculated based on distance - Right channel emphasis

UT-AUDIO-002: Volume Adjustment Based on Distance

Field	Details
Test ID	UT-AUDIO-002
Test Name	Adjust Audio Volume by Distance
Module	Spatial Audio Service
Priority	Medium
Preconditions	- Audio system initialized
Test Steps	1. Object at 5 meters: check volume 2. Object at 10 meters: check volume 3. Object at 20 meters: check volume
Test Data	Distances: 5m, 10m, 20m
Expected Result	- 5m: Volume = 80% - 10m: Volume = 60% - 20m: Volume = 40% - Inverse relationship with distance

UT-AUDIO-003: Audio Channel Panning

Field	Details
Test ID	UT-AUDIO-003
Test Name	Pan Audio to Correct Channel
Module	Audio Renderer
Priority	High
Preconditions	- Stereo audio output available
Test Steps	1. Position: LEFT → check left channel emphasis 2. Position: RIGHT → check right channel 3. Position: CENTER → check balance
Test Data	Positions: LEFT, RIGHT, CENTER

Expected Result	- LEFT: 80% left channel, 20% right - RIGHT: 20% left, 80% right - CENTER: 50% left, 50% right

2.5 Emergency Services Module

UT-EMG-001: Emergency Alert Message Creation

Field	Details
Test ID	UT-EMG-001
Test Name	Create Emergency Alert Message
Module	Emergency Service
Priority	Critical
Preconditions	- Emergency contacts configured - GPS location available
Test Steps	1. Trigger emergency alert 2. Acquire location 3. Generate message 4. Verify message content
Test Data	Location: (37.7749, -122.4194)
Expected Result	- Message contains: "EMERGENCY: User needs help" - Includes GPS coordinates - Includes timestamp - Includes location link

UT-EMG-002: Fall Detection Algorithm

Field	Details
Test ID	UT-EMG-002
Test Name	Detect Fall from Accelerometer Data
Module	Fall Detector
Priority	Critical
Preconditions	- Accelerometer active - Fall detection enabled
Test Steps	1. Simulate accelerometer data for fall: - Initial: (0, 0, 9.8) m/s^2 - During fall: (0, 0, 1.5) m/s^2 - Impact: (0, 0, 25) m/s^2 2. Process data 3. Check fall detection
Test Data	Acceleration pattern simulating fall

Expected Result	- Fall detected: TRUE - Detection latency < 1 second - Emergency alert triggered after 10-second countdown - User can cancel within countdown

UT-EMG-003: Emergency Contact Notification

Field	Details
Test ID	UT-EMG-003
Test Name	Send SMS to Emergency Contacts
Module	Alert Manager
Priority	Critical
Preconditions	- Emergency contacts: ["+1-555-0100", "+1-555-0200"] - SMS permission granted
Test Steps	1. Create emergency alert 2. Call sendToContacts() 3. Verify SMS sent
Test Data	Contacts: 2 emergency contacts
Expected Result	- SMS sent to both contacts - Each SMS contains location and message - Delivery confirmation received - Contacts notified count: 2

2.6 OCR Module

UT-OCR-001: Text Detection from Image

Field	Details
Test ID	UT-OCR-001
Test Name	Detect Text in Clear Image
Module	OCR Service
Priority	High
Preconditions	- OCR model loaded - Test image with clear text available
Test Steps	1. Load image: sign_test_01.jpg (contains "COFFEE SHOP") 2. Call scanImage(image) 3. Verify detected text
Test Data	Image containing: "COFFEE SHOP"

Expected Result	- Detected text: "COFFEE SHOP" - Confidence > 85% - Processing time < 1 second - Text bounding boxes returned

UT-OCR-002: Text Detection with Low Quality Image

Field	Details
Test ID	UT-OCR-002
Test Name	Handle Low Quality Image Gracefully
Module	OCR Service
Priority	Medium
Preconditions	- OCR service active
Test Steps	1. Load blurry/low-res image 2. Attempt text detection 3. Check confidence scores
Test Data	Blurry image
Expected Result	- Detected text (if any) has confidence < 60% - User notified: "Image quality too low" - Suggestion to improve lighting/focus - No crash or error

2.7 Offline Mode Module

UT-OFFLINE-001: Map Data Download

Field	Details
Test ID	UT-OFFLINE-001
Test Name	Download Map for Offline Use
Module	Offline Maps Service
Priority	High
Preconditions	- User logged in - Internet connection active - Storage space available
Test Steps	1. Select region: "San Francisco" 2. Call downloadMapData(region) 3. Monitor download progress
Test Data	Region: San Francisco (approx. 50MB)

Expected Result	- Download completes successfully - File saved to local storage - Map data integrity verified - User notified of completion

UT-OFFLINE-002: Offline Route Calculation

Field	Details
Test ID	UT-OFFLINE-002
Test Name	Calculate Route Using Cached Map
Module	Offline Navigation Engine
Priority	High
Preconditions	- Map data cached for area - No internet connection - GPS available (works offline)
Test Steps	1. Disable internet 2. Request route within cached area 3. Verify route calculation
Test Data	Start/End both in cached region
Expected Result	- Route calculated using local data - Navigation starts successfully - Turn-by-turn guidance works - No internet connection errors

2.8 Wearable Integration Module

UT-WEAR-001: Bluetooth Device Discovery

Field	Details
Test ID	UT-WEAR-001
Test Name	Discover Wearable Device via Bluetooth
Module	Bluetooth Manager
Priority	Medium
Preconditions	- Bluetooth enabled - Wearable device on and in pairing mode
Test Steps	1. Start Bluetooth scan 2. Wait for device discovery 3. Verify device appears in list
Test Data	Device: "SENSEI Haptic Band"

Expected Result	- Device discovered within 10 seconds - Device name: "SENSEI Haptic Band" - Signal strength displayed - Device is selectable

Implementation

1.12 Work Breakdown Structure (WBS)

Level 1: SENSEI App

1. **Requirement Analysis**
2. **UI/UX Design**
3. **AR Development**
4. **AI Integration**
5. **Haptic Wearable Integration**
6. **Database Setup**
7. **Testing & Feedback**
8. **Deployment**

1.13 Team Roles and Responsibilities

Table 5.1. Individual Tasks

Team Member	Activity
Neha Niamat	Frontend development, UI/UX design, functional requirements, testing
jaisha sohail	Backend development, database, API integration, architecture design, AR implementation

It is **MANDATORY** to provide a **GANTT CHART** to clearly illustrate the project schedule.

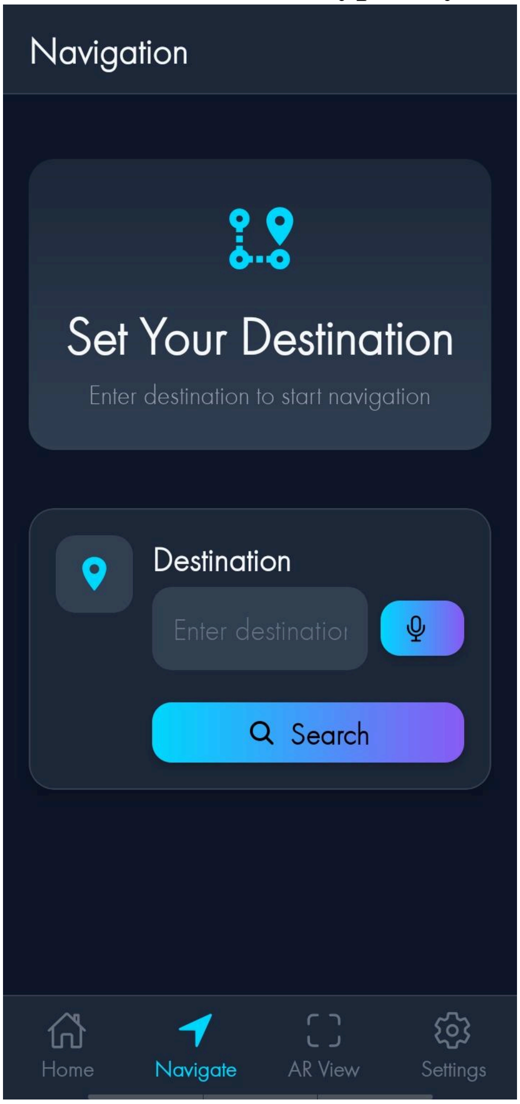
1.14 Tools and Technologies

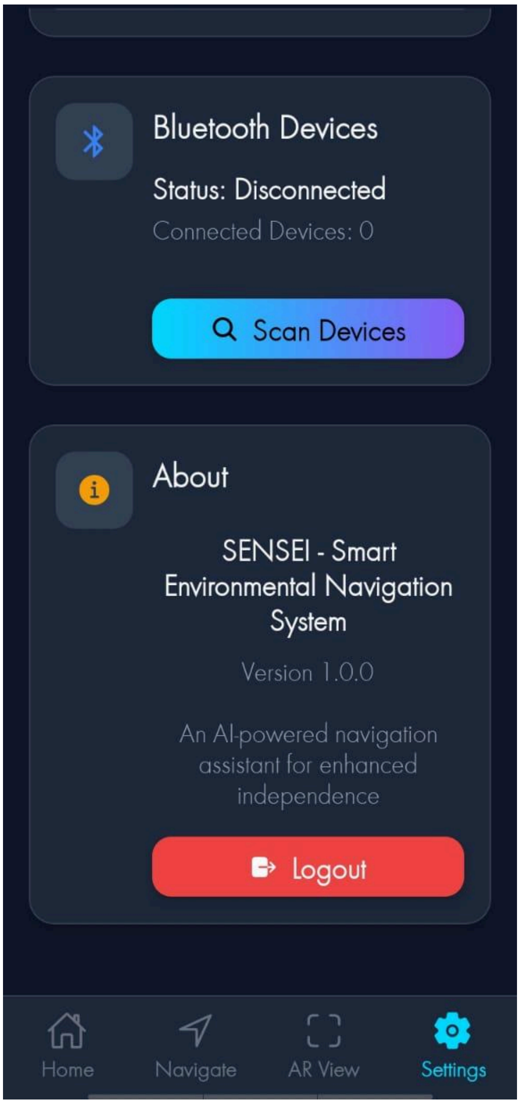
- **React Native (Expo)** : Mobile development
- **Expo AR / Three.js** : AR scene handling
- **TensorFlow Lite / PyTorch Mobile** : AI models
- **Expo Bluetooth API** : Wearable device integration
- **JavaScript** : Main programming language
- **Google Resonance Audio** : Spatial sound processing

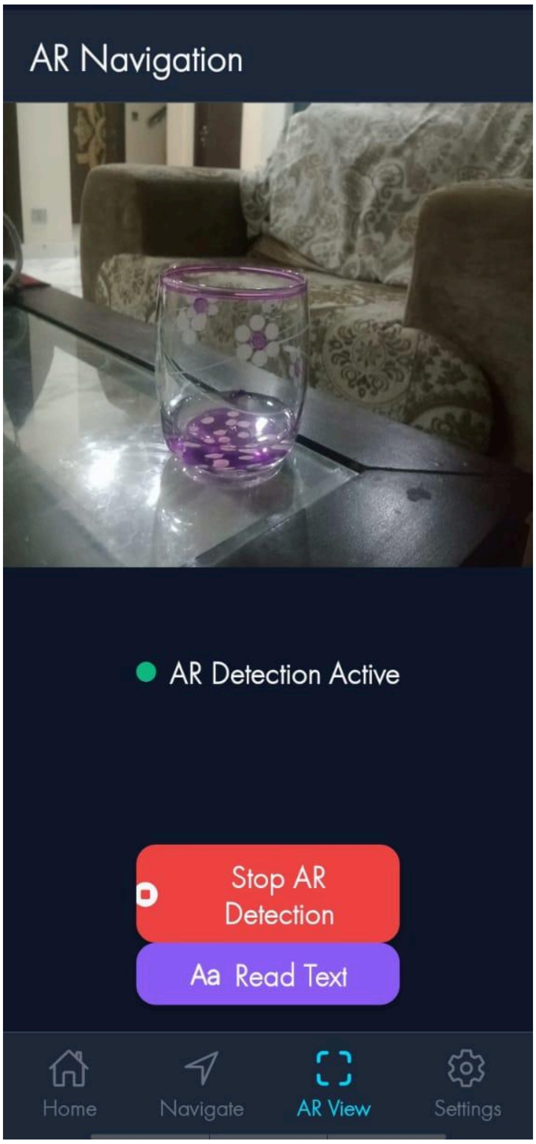
1.15 Implementation Details

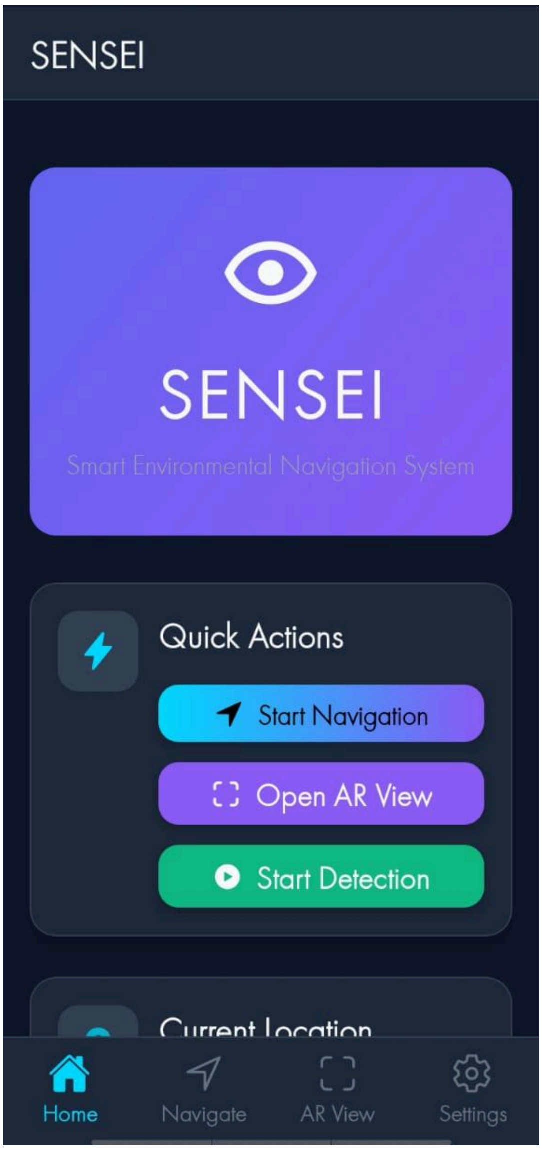
- AR scene created using Expo AR + Three.js to map surfaces.
- YOLO/TensorFlow Lite model used for object detection.
- Depth estimation applied for foot placement guidance.
- Emotion detection added via facial expression classifier.
- Navigation algorithm converts obstacle positions into spatial audio directions.
- Haptic feedback triggered using Bluetooth wearable patterns.
- Local storage used for saving route memory and settings.

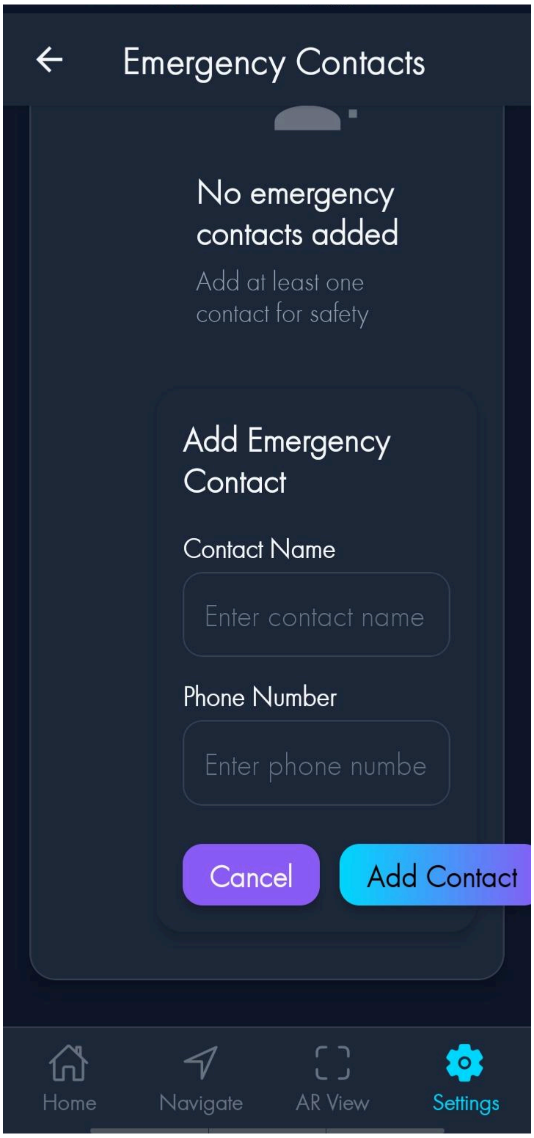
1.16 Screenshots of Prototype / System

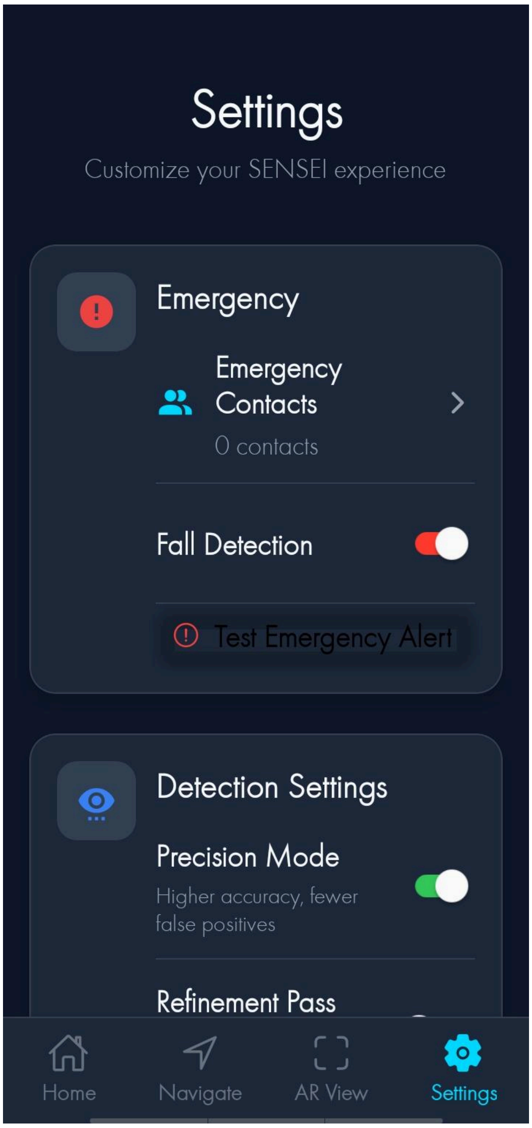


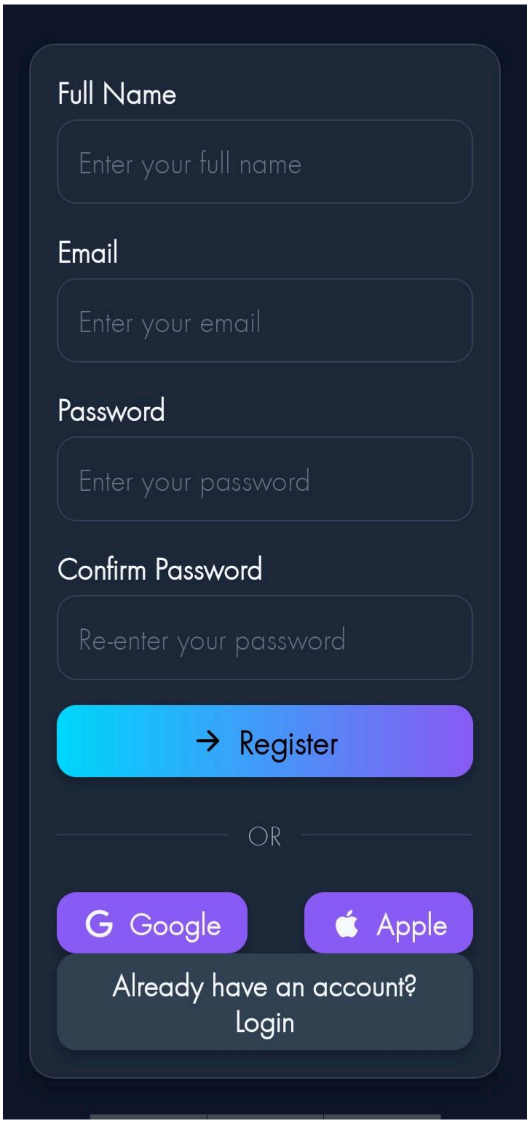
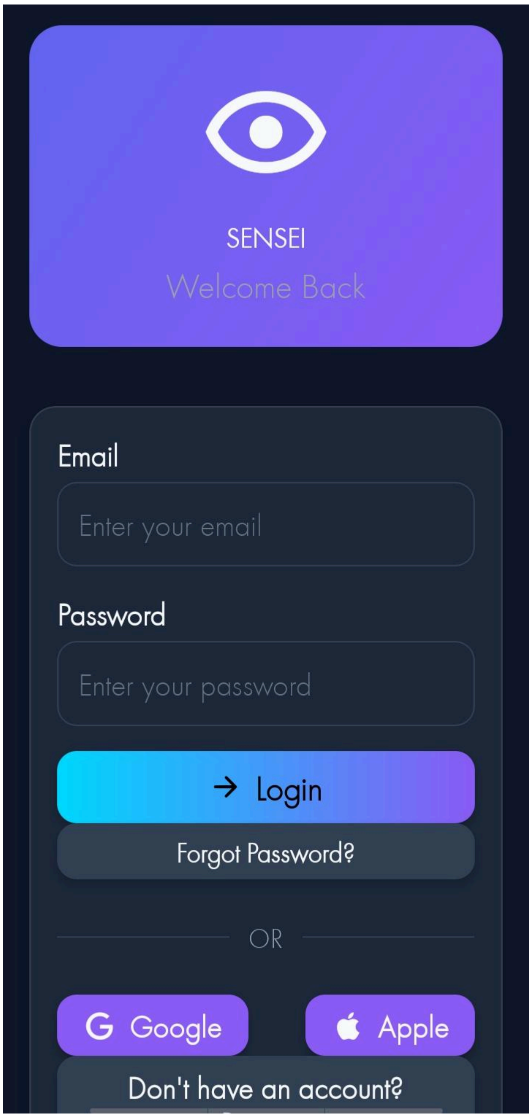


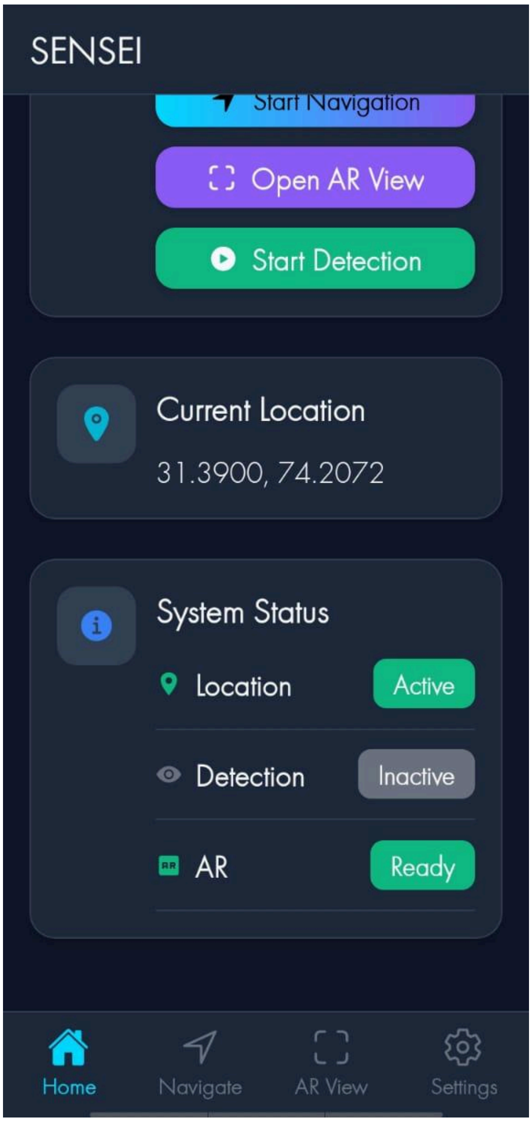

AR Navigation
AR Detection Active
Stop AR Detection
Aa Read Text
Home
Navigate
AR View
Settings


SENSEI
SENSEI
Smart Environmental Navigation System
Quick Actions
Start Navigation
Open AR View
Start Detection
Current Location
Home
Navigate
AR View
Settings


Emergency Contacts
No emergency contacts added
Add at least one contact for safety
Add Emergency Contact
Contact Name
Enter contact name
Phone Number
Enter phone numbe
Cancel
Add Contact
Home
Navigate
AR View
Settings


Settings
Customize your SENSEI experience
Emergency
Emergency Contacts
0 contacts
Fall Detection
Test Emergency Alert
Detection Settings
Precision Mode
Higher accuracy, fewer false positives
Refinement Pass
Home
Navigate
AR View
Settings


Full Name
Enter your full name
Email
Enter your email
Password
Enter your password
Confirm Password
Re-enter your password
Register
OR
Google
Apple
Already have an account?
Login

SENSEI
Welcome Back
Email
Enter your email
Password
Enter your password
Login
Forgot Password?
OR
Google
Apple
Don't have an account?



1.17 Challenges During Implementation

- Limited ARCore/ARKit access due to Expo restrictions.
- Mobile AI model optimization difficulties.
- Real-time processing frame rate drop in low-light environments.
- Testing with blind users requires controlled, supervised environments.
- Bluetooth wearable syncing issues.

References

1]
L. M. Lau, J. Curson, R. Drew, P. Dew, and C. Leigh, “Use Of VSP Resource Rooms to Support Group Work in a Learning Environment,” *Journal Paper Example*, vol. 99, pp. 1–2, 1999.

[2]
H. Nakanishi, C. Yoshida, T. Nishimura, and T. Ishida, "Freewalk: Supporting casual meetings in a network," in *Proc. 1996 ACM Conference on Computer Supported Cooperative Work*, Boston, Massachusetts, USA, 1996.

[3]
Google, "Lookout App – Object Detection and Accessibility Features," 2020. [Online]. Available: https://blog.google/technology/ai/lookout-app/. [Accessed: 23-Jun-2025].

[4]
Microsoft, "Seeing AI – A Talking Camera App for the Blind Community," 2024. [Online]. Available: https://www.microsoft.com/en-us/ai/seeing-ai. [Accessed: 23-Jun-2025].

[5]
TUAT Korea, "Sullivan+ – AI-Based Visual Assistance Application," 2019. [Online]. Available: https://www.mysullivan.org. [Accessed: 23-Jun-2025].

[6]
PR Newswire, "TUAT and LG Uplus Expand Sullivan+ Accessibility App Globally," 2019. [Online]. Available: https://www.prnewswire.com. [Accessed: 23-Jun-2025].

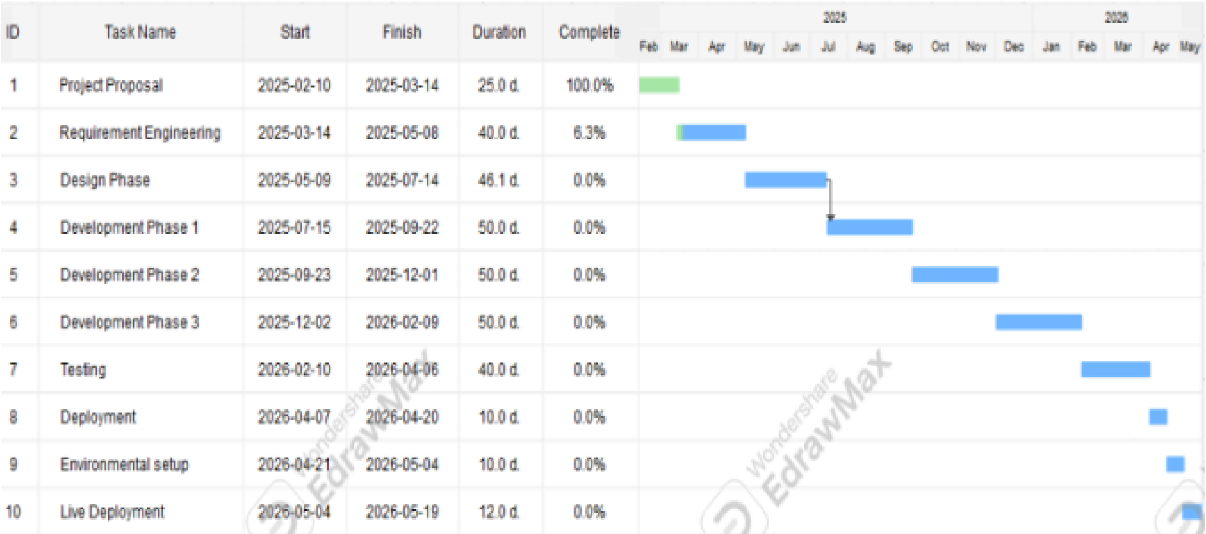
[7]
G. Coach, "How To Reference In Microsoft Word 2025: Full Tutorial With Examples Of Citations & Bibliography," YouTube, 2024. [Online]. Available: https://www.youtube.com/watch?v=wnviwr-CF2w. [Accessed: 23-Jun-2025].

[8]
Microsoft Inc., "Add citations in a Word document," Microsoft Support, 2024. [Online]. Available: https://support.microsoft.com/en-us/office/add-citations-in-a-word-document-ab9322bb-a8d3-47f4-80c8-63c06779f127. [Accessed: 23-Jun-2025].

[9]
Google Developers, "TensorFlow Lite – On-device Machine Learning," 2024. [Online]. Available: https://www.tensorflow.org/lite. [Accessed: 23-Jun-2025].

[10]
Expo Documentation, "Expo AR – Building Augmented Reality with React Native," 2024. [Online]. Available: https://docs.expo.dev. [Accessed: 23-Jun-2025].

GANTT CHART :



ID	Task Name	Start	Finish	Duration	Complete
1	Project Proposal	2025-02-10	2025-03-14	25.0 d.	100.0%
2	Requirement Engineering	2025-03-14	2025-05-08	40.0 d.	6.3%
3	Design Phase	2025-05-09	2025-07-14	46.1 d.	0.0%
4	Development Phase 1	2025-07-15	2025-09-22	50.0 d.	0.0%
5	Development Phase 2	2025-09-23	2025-12-01	50.0 d.	0.0%
6	Development Phase 3	2025-12-02	2026-02-09	50.0 d.	0.0%
7	Testing	2026-02-10	2026-04-06	40.0 d.	0.0%
8	Deployment	2026-04-07	2026-04-20	10.0 d.	0.0%
9	Environmental setup	2026-04-21	2026-05-04	10.0 d.	0.0%
10	Live Deployment	2026-05-04	2026-05-19	12.0 d.	0.0%

Neha Karamat

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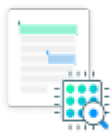

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